

claude-windows / 90c0b6b8-7430-4870-98cb-e487eacfa1d7

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\90c0b6b8-7430-4870-98cb-e487eacfa1d7\subagents\agent-ae31d85dcd8432b87.jsonl`  
- SHA-256: `374314c031dfbf8ee1880f26dc3726f98ecfdd2f0fb5515ff6888794d65e86a0`  
- Source modified: `2026-07-04T02:28:03+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `subagents`  
- session_id: `90c0b6b8-7430-4870-98cb-e487eacfa1d7`
```

Transcript

1. user (2026-07-04T02:26:08.292Z)

Repo root: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer.

I'm about to turn ON Cloudflare Access edge auth (milestone M9) for a DEPLOYED Cloud Run control-plane service (rally-control-plane, asia-southeast1), fronting it at api.khelsutra.guru. I need an exact, lockout-safe map. Read (don't modify) and report with file:line / doc-section evidence:

1. ****CF Access JWT verification**** ? read backend/api/auth.py fully. What request header(s) does it read (e.g. Cf-Access-Jwt-Assertion)? How does it verify the RS256 JWT (JWKS URL from cf_team_domain? audience == cf_access_aud?)? Is PyJWT[crypto] required (the `auth` extra)? Is it lazy-imported? What HTTP status on missing/invalid token? Which routes are gated vs. exempt (is /api/health exempt so Cloud Run startup/health probes still pass)?
2. ****Config keys + boot-time requirements**** ? for security.edge_auth_enabled, cf_team_domain, cf_access_aud, allowed_video_root (and any others under `security`): exact model field names + defaults (backend/config/models.py), and the exact startup checks (backend/config/startup.py ~100-140) that make the server sys.exit(1) at boot when edge_auth is ON. Enumerate EVERY field that MUST be set to avoid a boot crash. (I do NOT want to deploy a crash-looping revision.)
3. ****Bind host interaction**** ? confirm backend/server_runtime.py _resolve_bind_host and whether enabling edge_auth changes required flags (we currently pass --host 0.0.0.0 explicitly).
4. ****The `auth` pip extra**** ? is PyJWT[crypto] installed in the deploy image? Check deploy/cloudrun/Dockerfile ? the image installs `[storage-gcs]` (and my in-flight PR adds cloud-run). Is `auth` installed? If NOT, enabling edge_auth on the deployed image would fail JWT verify at runtime ? flag this as a probable gap and where PyJWT is imported.
5. ****Ingress lock + DNS/custom-domain**** ? search docs (docs/COMPUTE_DECOUPLED_SERVING/README.md, any M9 doc, deploy/cloudrun/README.md, docs/ for "M9", "edge_auth", "Cloudflare", "Access", "api.khelsutra.guru", "ingress", "domain mapping") for the DOCUMENTED procedure to (a) lock Cloud Run ingress so only Cloudflare reaches the origin, and (b) map api.khelsutra.guru to the service (Cloud Run domain mapping vs. a load balancer vs. CF proxy to the run.app URL). Quote the exact documented steps/commands if present.
6. ****Existing Access app to mirror**** ? any repo notes/docs referencing the existing "KhelSutra Studio (alpha)" Cloudflare Access application (on app.khelsutra.guru) whose config (team domain, AUD extraction) I should mirror for api.khelsutra.guru.

Return a concise structured summary: (A) exact config JSON I must add for edge auth (all required keys), (B) whether the deployed image can verify JWTs as-is or needs the auth extra (image rebuild), (C) the documented ingress-lock + DNS approach, (D) any boot-crash traps. Cite file:line throughout.

2. assistant (2026-07-04T02:26:10.320Z)

I'll conduct a thorough read-only exploration of the codebase to map the Cloudflare Access edge auth implementation. Let me start by examining the key files systematically.

3. user (2026-07-04T02:26:10.697Z)

```
backend\eval\__init__.py
backend\utils\__init__.py
backend\tools\__init__.py
backend\eval\audio\__init__.py
backend\pipeline\audio\__init__.py
backend\features\__init__.py
backend\utils\hashing.py
backend\pipeline\personalization\__init__.py
backend\api\__init__.py
backend\config\__init__.py
backend\pipeline\segmenters\__init__.py
backend\pipeline\validators\__init__.py
backend\results.py
backend\sports\base.py
backend\utils\redact.py
backend\worker\__init__.py
backend\eval\gemini_refine.py
backend\annotations\automated_provider.py
backend\annotations\base.py
backend\annotations\file_provider.py
backend\annotations\__init__.py
backend\api\auth.py
backend\billing.py
backend\compute\base.py
backend\compute\__init__.py
backend\config\loader.py
backend\eval\annotations.py
backend\eval\audio\e1_probe.py
backend\eval\batch.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\classifier.py
backend\eval\eval_partial_labels.py
backend\eval\eval_stats.py
backend\eval\experiment.py
backend\eval\export_wasb_segments.py
backend\eval\fusion_compare.py
backend\eval\golden_fixtures.py
backend\eval\golden_real_fixtures.py
backend\eval\gt_loader.py
backend\eval\harness.py
backend\eval\metrics.py
backend\eval\per_user_eval.py
backend\eval\per_user_gate.py
backend\eval\per_user_regression.py
backend\eval\promotion.py
backend\eval\regression.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
backend\eval\splits.py
backend\eval\tolerance_metrics.py
backend\eval\training.py
backend\features\rally_seq.py
backend\flywheel.py
backend\infrastructure\execution_policy.py
backend\pipeline\detectors\audio_features.py
backend\pipeline\detectors\__init__.py
backend\pipeline\detectors\court_gate.py
backend\pipeline\detectors\device.py
backend\pipeline\detectors\fusion_features.py
```

backend\pipeline\detectors\person_detector.py
backend\pipeline\detectors\stub_runner.py
backend\pipeline\personalization\serving.py
backend\pipeline\segmenters\base.py
backend\pipeline\segmenters_keyhint.py
backend\pipeline\segmenters\fusion_hybrid.py
backend\pipeline\segmenters\motion.py
backend\pipeline\segmenters\tracknet_hybrid.py
backend\pipeline\segmenters\wasb_hybrid.py
backend\pipeline\stitcher\watermark_i18n.py
backend\pipeline\validators\base.py
backend\pipeline\validators\blur.py
backend\pipeline\validators\relevance.py
backend\pipeline\validators\scene_cut.py
backend\providers\base.py
backend\providers\gemini.py
backend\providers__init__.py
backend\reporting\harvest.py
backend\reporting__init__.py
backend\reporting\run_signals.py
backend\sports__init__.py
backend\sports\badminton.py
backend\storage\base.py
backend\storage\gdrive.py
backend\storage\local.py
backend\storage\ref.py
backend\storage\s3.py
backend\tools\export.py
backend\tools\rally_slicer.py
backend\utils\app_home.py
backend\utils\geometry.py
backend\utils\logging_setup.py
backend\utils\paths.py
backend\utils\run_telemetry.py
backend\utils\single_instance.py
backend\utils\workspace.py
backend\eval>windowing.py
backend\pipeline\detectors\rally_gate.py
backend\pipeline\segmenters\trajectory_hybrid.py
backend\pipeline\detectors\base.py
(Showing 100 of 144 matching files; 44 more are not listed. Narrow the pattern or path to see the rest.)

4. user (2026-07-04T02:26:10.922Z)

deploy\cloudrun\cloudbuild.yaml
deploy\digitalocean\gpu_setup.sh
deploy\digitalocean\mount_scratch.sh
deploy\gcp\nightly_regression\Dockerfile_launcher
deploy\gcp\nightly_regression\cloudbuild_launcher.yaml
deploy\gcp\nightly_regression\run_on_vm.sh
deploy\gcp\ops-agent-startup.sh
deploy\digitalocean\README.md
deploy\gcp\buckets.env
deploy\gcp\create_gpu_vm.sh
deploy\gcp\nightly_regression\create_nightly_bucket.sh
deploy\gcp\nightly_regression\README.md
deploy\gcp\README.md
deploy\gcp\nightly_regression\launch.sh
deploy\gcp\nightly_regression\register_scheduler.sh
deploy\gcp\nightly_regression\setup_email.sh
deploy\gcp\provision.sh
deploy\gcp\teardown.sh
deploy\digitalocean\bootstrap.sh
deploy\cloudrun\Dockerfile

deploy\cloudrun\README.md

5. user (2026-07-04T02:26:11.135Z)

docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\mockups\ui_alpha_mockup.html
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\ABLATION_LAYER.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\EVALUATION.md
docs\EXECUTION_POLICY.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\PERSONALIZATION_PLAN.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\STORAGE_SHARING_MODEL.md
docs\TRACKNET_WSL_SETUP.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\VIDEO_LOCALITY_MODEL.md
docs\CLOUD_GPU_DRYRUN_GUIDE.md
docs\COMMERCIALIZATION.md
docs\COMPUTE_DECOUPLED_SERVING\README.md
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md
docs\COMPUTE_DECOUPLED_SERVING\architecture.svg
docs\COMPUTE_DECOUPLED_SERVING\runlogs\README.md
docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_truly-local_TEMPLATE.md
docs\CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md
docs\CREATING_REELS.md
docs\EXPERIMENT_HARNESS.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\PACKAGING_PLAN.md
docs\PER_VIDEO_WORKSPACE.md
docs\PROJECT_DOC_TEMPLATE.md
docs\PROMINENT_COURT_DETECTION.md
docs\R1_MILESTONE_LAUNCH_REPORT.md
docs\R2_EVAL_METRIC_DESIGN.md
docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\RALLY_DETECTION_QUALITY_REPORT.md
docs\RALLY_QUALITY_RESEARCH.md
docs\REAL_FOOTAGE_VALIDATION_REPORT.md
docs\ROADMAP.md
docs\SETUP_NEW_MACHINE.md
docs\UI_PROPOSAL.md
docs\archives\PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md
docs\archives\decisions\DECISIONS.md

docs\archives\field-pmf\FIELD_VISIT_ANSWER_SHEET.md
docs\archives\field-pmf\FIELD_VISIT_PLAYBOOK.md
docs\archives\field-pmf\PMF_FIELD_SIGNALS.md
docs\archives\field-pmf\PMF_MARKET_RESEARCH.md
docs\archives\field-pmf\PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs\archives\field-pmf\README.md
docs\archives\past_projects\DECOUPLED_COMPUTE\01-EXISTING-PLANS-AND-GAP.md
docs\archives\past_projects\DECOUPLED_COMPUTE\02-COMPUTE-MAP-AND-CONTRACT.md
docs\archives\past_projects\DECOUPLED_COMPUTE\03-PLATFORM-OPTIONS.md
docs\archives\past_projects\DECOUPLED_COMPUTE\04-BUILD-AND-TEST-PLAN.md
docs\archives\past_projects\DECOUPLED_COMPUTE\05-COST-ATTRIBUTION.md
docs\archives\past_projects\DECOUPLED_COMPUTE\06-RISKS-AND-OPEN-QUESTIONS.md
docs\archives\past_projects\DECOUPLED_COMPUTE\07-COMPUTE-ABSTRACTION.md
docs\archives\past_projects\DECOUPLED_COMPUTE\DEPLOY_REPORT_GCP_2026-06-20.md
docs\archives\past_projects\DECOUPLED_COMPUTE\HANDOFF.md
docs\archives\past_projects\DECOUPLED_COMPUTE\README.md
docs\archives\past_projects\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\README.md
docs\archives\roora-vendor\GOLDEN_SET_VENDORING.md
docs\archives\roora-vendor\README.md
docs\archives\roora-vendor\ROORA_LABELING_GUIDELINES.md
docs\data_pipeline\GOLDEN_DATA_SHARING.md
docs\data_pipeline\GOLDEN_SET_IMPLEMENTATION.md
docs\data_pipeline\GOLDEN_SET_PHASE1_VIDEOS.md
docs\data_pipeline\README.md
docs\KHELSUTRA_PRODUCT_OVERVIEW.md
docs\DECISION_SNAPSHOT_REGRESSION.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py
docs\NIGHTLY_ENGINE_REGRESSION.md
docs\RALLY_DETECTION_CODE_REVIEW.md
docs\data_pipeline\VIDEO_RECORDING_GUIDELINES.md
docs\I18N_PLAN.md
docs\QUALITY_ITERATIONS.md
docs\RALLY_DETECTION_FIXES_PLAN.md
docs\DESKTOP_APP_PLAN.md
docs\PLATFORM_ARCHITECTURE.md
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md
(Showing 100 of 111 matching files; 11 more are not listed. Narrow the pattern or path to see the rest.)

6. assistant (2026-07-04T02:26:12.702Z)

Now let me read the critical files directly:

7. user (2026-07-04T02:26:13.370Z)

```
1      """Cloudflare Access edge-auth ? verify the Cf-Access JWT, derive the tenant ``user_id``
(M9, D9).
2
3      **DEFAULT-OFF:** when ``security.edge_auth_enabled`` is False the resolver is a strict
NO-OP (returns
4      ``None`` = the local single-user) ? nothing about the existing local path changes. When
enabled, the
5      ``Cf-Access-Jwt-Assertion`` header is **required** and **verified** (RS256) against the
team JWKS ?
6      signature + exp + issuer + audience ? so a direct hit to the Cloud Run URL **can't
spooft** the
7      identity header (the explicit D9 anti-spoof requirement). Returns the verified
``user_id`` (the
8      token's ``email``, else ``sub``).
9
10     ``jwt`` (``PyJWT[crypto]``) is **lazy-imported** so the base runtime works without it
(install the
```

```

11     ``auth`` extra for hosted/multi-tenant). The JWKS fetch is **injectable**, so the whole
verifier is
12     unit-tested **offline** with an in-test RSA keypair + a fake JWKS ? no network, no
Cloudflare.
13     """
14
15     from __future__ import annotations
16
17     import json
18     import time
19     import urllib.request
20     from typing import Any, Callable, Mapping, Optional
21
22     # Cloudflare Access sends the signed assertion in this header (it terminates at the CF
edge).
23     CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
24     _JWKS_PATH = "/cdn-cgi/access/certs"
25     _JWKS_TTL_SEC = (
26         600.0 # re-fetch the team JWKS at most every 10 min (keys rotate slowly)
27     )
28
29     JwksFetcher = Callable[[str], dict]
30
31
32     class EdgeAuthError(Exception):
33         """A required Cf-Access token was missing or failed verification ? the caller
returns 401/403."""
34
35
36     # module-level JWKS cache: team_domain -> (fetched_at, jwks_dict). Production only;
tests inject.
37     _jwks_cache: dict[str, tuple[float, dict]] = {}
38
39
40     def _issuer(team_domain: str) -> str:
41         return team_domain.rstrip("/")
42
43
44     def _default_jwks_fetcher(team_domain: str) -> dict:
45         """Fetch the team's public JWKS (`<team_domain>/cdn-cgi/access/certs`). Network ?
only the
46         production path reaches here; tests pass an injected fetcher."""
47         url = _issuer(team_domain) + _JWKS_PATH
48         with urllib.request.urlopen(url, timeout=5) as resp: # noqa: S310 (https team
domain)
49             data: Any = json.load(resp)
50             if not isinstance(data, dict) or "keys" not in data:
51                 raise EdgeAuthError(f"unexpected JWKS shape from {url}")
52             return data
53
54
55     def _get_jwks(team_domain: str, fetcher: Optional[JwksFetcher]) -> dict:
56         """Return the JWKS for ``team_domain``, cached with a TTL. An injected ``fetcher``
(tests)
57         bypasses the cache so each test is deterministic."""
58         if fetcher is not None:
59             return fetcher(team_domain)
60         now = time.time()
61         cached = _jwks_cache.get(team_domain)
62         if cached and (now - cached[0]) < _JWKS_TTL_SEC:
63             return cached[1]
64         jwks = _default_jwks_fetcher(team_domain)
65         _jwks_cache[team_domain] = (now, jwks)
66         return jwks
67

```

```

68
69     def verify_cf_access_token(
70         token: str, *, team_domain: str, aud: str, jwks: dict
71     ) -> str:
72         """Verify a Cf-Access JWT against ``jwks`` (RS256; checks sig + exp + issuer +
audience) and
73         return the user_id (``email`` else ``sub``). Raises ``EdgeAuthError`` on ANY
failure."""
74         try:
75             import jwt
76             from jwt import PyJWKSet
77         except (
78             Exception
79         ) as exc: # pragma: no cover - only when the auth extra isn't installed
80             raise EdgeAuthError(
81                 "PyJWT[crypto] is required for edge auth ? install the 'auth' extra "
82                 "(pip install -e .[auth]) on a host with edge_auth_enabled."
83             ) from exc
84
85         try:
86             kid = jwt.get_unverified_header(token).get("kid")
87             if not kid:
88                 raise EdgeAuthError("token header has no kid")
89             signing_key = PyJWKSet.from_dict(jwks)[kid]
90             claims = jwt.decode(
91                 token,
92                 signing_key.key,
93                 algorithms=["RS256"],
94                 audience=aud,
95                 issuer=_issuer(team_domain),
96                 options={"require": ["exp", "iss", "aud"]},
97             )
98         except EdgeAuthError:
99             raise
100     except KeyError as exc: # kid not in the JWKS
101         raise EdgeAuthError(f"no JWKS key for kid {exc}") from exc
102     except (
103         Exception
104     ) as exc: # jwt.* errors: bad sig / expired / wrong aud|iss / malformed
105         raise EdgeAuthError(f"Cf-Access token verification failed: {exc}") from exc
106
107     user = claims.get("email") or claims.get("sub")
108     if not user:
109         raise EdgeAuthError("verified token carries no email/sub claim")
110     return str(user)
111
112
113     def resolve_tenant(
114         headers: Mapping[str, str],
115         config: Any,
116         *,
117         jwks_fetcher: Optional[JwksFetcher] = None,
118     ) -> Optional[str]:
119         """Resolve the tenant ``user_id`` for a request, or ``None`` when edge auth is OFF.
120
121         DEFAULT-OFF (``security.edge_auth_enabled=False``) ? ``None`` (the local single-
user; unchanged).
122         Enabled ? require + verify the Cf-Access header; raise ``EdgeAuthError`` (? 401/403)
on any miss.
123         ``jwks_fetcher`` is injectable for offline tests.
124         """
125         sec = getattr(config, "security", None)
126         if not sec or not getattr(sec, "edge_auth_enabled", False):
127             return None # DEFAULT-OFF ? no tenant identity, byte-identical to today
128

```

```

129         # Strip so a whitespace-only value counts as unset ? matching the boot-time exposure
check in
130         # config/startup.py (EXPOSED_EDGE_AUTH_INCOMPLETE); one semantics for "configured",
both places.
131         team_domain = (getattr(sec, "cf_team_domain", "") or "").strip()
132         aud = (getattr(sec, "cf_access_aud", "") or "").strip()
133         if not team_domain or not aud:
134             raise EdgeAuthError(
135                 "edge_auth_enabled but security.cf_team_domain / cf_access_aud are not
configured"
136             )
137
138         # Headers are case-insensitive over HTTP; Starlette's Headers handles that, but
support a plain
139         # dict too (tests) by trying both casings.
140         token = headers.get(CF_ACCESS_HEADER) or headers.get(CF_ACCESS_HEADER.lower())
141         if not token:
142             raise EdgeAuthError(
143                 f"missing {CF_ACCESS_HEADER} header (request did not pass Cloudflare
Access)"
144             )
145
146         jwks = _get_jwks(team_domain, jwks_fetcher)
147         return verify_cf_access_token(token, team_domain=team_domain, aud=aud, jwks=jwks)
148

```

8. user (2026-07-04T02:26:14.160Z)

```

1     """Pydantic models for the application configuration.
2
3     This module defines the canonical, validated, typed configuration for the
4     Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6     Unknown / typo'd config keys are rejected at load time via extra="forbid".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         pts_max_gap_seconds: float = Field(
31             default=10.0,
32             ge=0.0,
33             description=(
34                 "Maximum allowed keyframe gap (seconds) before flagging as spliced. "
35                 "GoPro H.264/H.265 recordings typically have GOP intervals of 1?4s; "
36                 "a gap >10s suggests edited/missing content. Mobile phones, screen "
37                 "recordings, and other encoders may use different GOP sizes ? tune "

```



```

38         "this threshold for your recording device."
39     ),
40 )
41 relevance: ValidationRelevanceConfig = Field(
42     default_factory=ValidationRelevanceConfig
43 )
44
45
46 class TrackNetConfig(BaseModel):
47     wsl_distro: str = "Ubuntu"
48     repo_dir: str = "~/models/TrackNetV4"
49     conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
50     conda_env: str = "TrackNetV4"
51     weights_path: str = ""
52     queue_length: int = 5
53     stage_in_wsl: bool = True
54     wsl_stage_dir: str = "~/clips"
55     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
56     velocity_threshold: float = 0.02
57     # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
58     # decoded frame cache) are deleted automatically ? they are pure intermediates,
59     # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
60     # true only for debugging. On FAILURE they are always kept so a re-run can resume.
61     keep_frames: bool = False
62     # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
the
63     # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
not warns).
64     # 0 = no extra floor (the headroom guard still applies).
65     min_free_gb: float = 0.0
66
67
68 class WasbIndexConfig(BaseModel):
69     """Typed config for the live WASB shuttle substrate (indexing.wasb).
70
71     Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
72     these knobs actually take effect ? previously this whole block was silently dropped
73     because nested config sections defaulted to extra='ignore'.
74     """
75
76     wsl_distro: str = "Ubuntu"
77     repo_dir: str = "~/models/WASB-SBDT"
78     conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
79     conda_env: str = "wasb"
80     weights_path: str = (
81         "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
82     )
83     sport: str = "badminton"
84     wsl_stage_dir: str = "~/clips"
85     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
86     velocity_threshold: float = 0.02
87     # --- Linux-native runner only (detector_impl='native'); ignored by the WSL runner.
---
88     # Env vars override these (WASB_DEVICE / WASB_PYTHON) for a config-free GPU box.
89     # torch device for the native runner. "auto" = cuda-if-available-else-cpu (the M4
CPU fallback
90     # for a no-GPU box); "cuda" stays strict (hard-fails without a GPU ? no silent slow-
CPU downgrade).
91     device: str = "cuda" # auto | cuda | mps | cpu
92     python_bin: str = (
93         "python" # the `wasb` conda env python (invoked by path, no `conda activate`)
94     )
95     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
96     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable

```

```

97         # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
98         keep_frames: bool = False
99         # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
the
100         # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
not warns).
101         # 0 = no extra floor (the headroom guard still applies).
102         min_free_gb: float = 0.0
103         # FAST PATH ? decode the video on the fly and feed frames straight to the detector,
104         # skipping the slow per-frame PNG extraction that dominates a long clip (~46k PNGs
for a
105         # 12-min 1080p60 video, which overran the 30-min timeout). Output is bit-identical
to the
106         # PNG path (PNG is lossless) ? VERIFIED on a real GPU 2026-06-20 (native stream ==
WSL disk,
107         # 1194/1195 frames byte-identical; native run-to-run byte-identical /
deterministic).
108         # Tri-state:
109         #     None          = per-runner default ? the WSL runner keeps DISK extraction
(unchanged Windows
110         #                     behaviour); the Linux-native runner (GPU box) defaults to STREAMING
(the perf win).
111         #     True/False = force on/off for BOTH runners (e.g. set False for a container cv2
can't seek).
112         stream_video: Optional[bool] = None
113         # GPU-feed tuning for the native runner (measured on the first real L4 serving run,
114         # 2026-07-03: the default batch 8 with single-process loading left the GPU <1%
utilized ?
115         # the pipeline was CPU-feed-bound). None = wasb_infer.py's parity defaults (batch 8,
no
116         # prefetch ? byte-identical to the historical path). The cloud-serving preset opts
into
117         # the measured fast values; local/WSL behaviour is unchanged unless set explicitly.
118         batch_size: Optional[int] = None
119         # Depth of the bounded producer-thread queue that overlaps CPU-side batch assembly
120         # (decode + PIL + ToTensor) with GPU inference in streaming mode. None/0 = off.
121         prefetch_batches: Optional[int] = None
122
123         @field_validator("batch_size")
124         @classmethod
125         def _batch_size_positive(cls, v: Optional[int]) -> Optional[int]:
126             # A DataLoader batch must be >=1; reject at config time so a typo can't reach
127             # `wasb_infer.py --batch-size` as an invalid value (the runner only passes the
flag
128             # when truthy, so 0 stays "off/parity-default").
129             if v is not None and v < 1:
130                 raise ValueError(f"indexing.wasb.batch_size must be >= 1 when set (got
{v})")
131             return v
132
133         @field_validator("prefetch_batches")
134         @classmethod
135         def _prefetch_batches_non_negative(cls, v: Optional[int]) -> Optional[int]:
136             # 0 = off (the default). Reject negatives: the runner passes any truthy value as
137             # `--prefetch-batches`, and `_PrefetchIter` clamps depth with max(1,
int(depth)), so a
138             # negative would SILENTLY enable prefetch at depth 1 instead of failing loudly.
139             if v is not None and v < 0:
140                 raise ValueError(
141                     f"indexing.wasb.prefetch_batches must be >= 0 when set (got {v})"
142                 )
143             return v
144
145

```

```

146 class FusionConfig(BaseModel):
147     """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
148
149     Every default reproduces control behaviour: the fusion segmenter persists the
150     shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
151     true ? the person-cue features, but it does NOT gate the served handoff unless a
152     threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court
mask
153     is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-
only);
154     supplied ? off-court persons (spectators / adjacent courts) are excluded.
155     """
156
157     # Which trajectory substrate seeds the windows (shuttle stays primary).
158     substrate: Literal["wasb", "tracknet"] = "wasb"
159     # Windowing velocity threshold for the fusion family (the engine reads
160     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
161     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
162     velocity_threshold: float = 0.02
163     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
164     # enabled ? so the default path never imports torch / touches the GPU.
165     cue_flags: dict[str, bool] = Field(default_factory=dict)
166     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
167     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
168     min_crossings: int = Field(default=0, ge=0)
169     # Served Gate B: when the person cue is on, drop windows with median in-court
players
170     # < this. 0 = OFF.
171     min_players: int = Field(default=0, ge=0)
172     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
173     court_polygon: Optional[list[list[float]]] = None
174     # Net line for the person two-sided-motion split (person features only ? the shuttle
175     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
176     net_axis: Literal["x", "y"] = "y"
177     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
178
179
180 class IndexingConfig(BaseModel):
181     default_segmenter: str = "yolo_hybrid"
182     use_gpu: bool = True
183     # Detector backend for the trajectory hybrids (wasb/tracknet). "auto" = the real
184     # WSL runner (default, production); "stub" = a deterministic CPU mock (no GPU/WSL)
185     # so the whole pipeline is regression-testable on a CPU box / CI; "native" = the
186     # Linux-native WASB runner (no WSL/conda-activate; a GPU box ? M3.5). The
187     # RALLY_DETECTOR_IMPL env var overrides this. See
pipeline/detectors/{stub,native_wasb}_runner.py.
188     detector_impl: str = "auto"
189     motion_threshold: float = 0.08
190     min_segment_duration: float = 3.0
191     dead_time_merge_gap: float = 2.0
192     chunk_duration: float = 90.0
193     chunk_overlap: float = 10.0
194     detector_model: str = "fasterrcnn_resnet50_fpn"
195     detector_score_threshold: float = 0.5
196     yolo_velocity_threshold: float = 0.15
197     yolo_frame_skip: int = 3
198     yolo_tracker: str = "bytetrack.yaml"
199     yolo_prefetch_workers: int = 2
200     min_action_window_duration: float = 2.0
201     # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
= a window

```

```

202      # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
203      # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't
swallow
204      # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
205      # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13
206      # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
207      # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
208      max_window_duration: float = 6.0
209      window_min_split_gap: float = (
210          0.5 # min internal gap (s) that qualifies as a split point
211      )
212      # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
213      # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
214      # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
215      # Only cuts (recall can't drop). OFF by default until measured/promoted.
216      window_hard_cap: bool = False
217      # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
218      # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
219      # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
220      # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
221      # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
222      # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
223      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
224      window_inactivity_end: float = 1.5
225      inactivity_rally_thresh: float = 0.20
226      inactivity_min_density: float = 0.30
227      # R4 density DENOMINATOR fix (code-review finding B). The R4 trim's fast-fraction
denominator is
228      # the COUNT of active samples in the trailing window, which sparse tracking inflates
(one fast
229      # sample after a gap ? 100% dense ? tail held open). When True, the denominator is
the window's
230      # TEMPORAL length instead. No-op under dense tracking (sample-count == temporal-
span), so it only
231      # corrects the sparse-track case. *** P2b PROMOTED DEFAULT-ON (2026-06-29): golden-
set A/B at the
232      # SERVED operating point (n=15, 595 rallies) ? tolF1 0.353?0.364 (?tolF1 +0.010,
sign 10W/4L, NO
233      # regression), |?end| 2.97?2.72s, seg_count_ratio ?0.103 (p=0.006), split_rate
?0.010 (p=0.016),
234      # R1 over-seg guard PASS; served_regresses(eps=0.0)=False ?
promote_if_served_safe(structural)=True.
235      # See docs/RALLY_DETECTION_FIXES_PLAN.md §7/§8. ***
236      inactivity_temporal_density: bool = True
237      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
238      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
239      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.

```

```

240         # mahadevpura-1's ~65s residual ? 11x a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
241         # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
242         # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
243         # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
244         # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
245         window_blob_fallback: bool = False
246         # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
247         # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
248         # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
249         # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
250         window_blob_factor: float = 4.0
251         log_yolo_candidates: bool = True
252         store_yolo_candidates: bool = True
253         ai_handoff_padding: float = 2.0
254         ai_max_workers: int = 5
255         # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
256         # chunks of a high-bitrate (4K) source blow past the provider upload cap
257         # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
258         # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
259         # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
260         # Matches gemini_refine's --clip-scale-width default.
261         ai_chunk_scale_width: int = 1280
262         # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
263         # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
264         # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
265         # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
266         # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
267         skip_ai_handoff: bool = False
268         # Optional debug knob: trim every video to the first N seconds before processing.
269         debug_duration: Optional[float] = None
270
271         tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
272         wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
273         fusion: FusionConfig = Field(default_factory=FusionConfig)
274
275         @field_validator("default_segmenter")
276         @classmethod
277         def segmenter_must_be_registered(cls, v: str) -> str:
278             # Lazy import: the segmenter registry imports modules that may import
279             # config, so we resolve it at validation time rather than at module load.
280             from backend.pipeline.segmenters.base import segmenter_registry
281
282             if segmenter_registry.get(v) is None:
283                 available = sorted(segmenter_registry.list_available().keys())
284                 raise ValueError(
285                     f"Segmenter '{v}' is not registered. Available: {'', ' '.join(available)}"
286                 )
287             return v
288
289         @model_validator(mode="after")
290         def _chunk_step_must_be_positive(self) -> "IndexingConfig":
291             # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive

```

```

292         # step makes `while t < duration: t += step` loop forever, growing memory before
any
293         # GPU work. Reject the bad config at load time instead.
294         if self.chunk_overlap >= self.chunk_duration:
295             raise ValueError(
296                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
297                 f"({self.chunk_duration}); otherwise chunking never advances."
298             )
299         return self
300
301
302     class OutputConfig(BaseModel):
303         default_highlights_name: str = "highlights.mp4"
304         db_name: str = "sports_indexer.db"
305         # Directory where the DB, highlight reels, and processed clips are written.
306         # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
307         output_dir: str = "output"
308
309
310     class WatermarkConfig(BaseModel):
311         """Branding overlay burned into each highlight clip during the per-clip re-encode
312         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
313         (under `stitching.watermark` in config.json) to turn it off. The text is fully
314         configurable. The concat stage stays stream-copy (-c copy)."""
315
316         enabled: bool = True
317         text: str = "Generated by Khelsutra.guru"
318         logo_path: str = "" # optional transparent PNG overlay; "" = no logo
319         font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family below
320         font: str = "Sans" # fontconfig family used when font_path is empty
321         font_size_ratio: float = Field(
322             default=0.035, gt=0.0, le=0.5
323         ) # text height as a fraction of frame height
324         opacity: float = Field(default=0.85, ge=0.0, le=1.0)
325         box: bool = True # faint backing box behind the text for legibility
326         margin: int = Field(default=24, ge=0) # px inset from the chosen corner
327         position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = (
328             "bottom-right"
329         )
330
331
332     class StitchingConfig(BaseModel):
333         begin_padding: float = 0.5
334         end_padding: float = 0.5
335         use_cache: bool = False
336         min_shot_count: int = 1
337         # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
338         # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
339         # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
340         # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
341         # local detector over-fires and its false positives are mostly SHORT (motion blips,
342         # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
343         # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
344         min_rally_duration: float = Field(default=0.0, ge=0.0)
345         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
346
347
348     class LoggingConfig(BaseModel):
349         """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""

```

```

350
351 # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
352 # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
353 # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
354 # them to WARNING; set true to restore full chatter for request-flow debugging.
355 verbose_http: bool = False
356
357
358 class SecurityConfig(BaseModel):
359     # When set, /api/process video_path must resolve under this directory (hosted/tenant
360     # mode). Default None preserves the single-user local 'any local file' workflow.
361     allowed_video_root: Optional[str] = None
362
363     # --- Chunked upload (B2, PR-2) ---
364     # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
365     # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
366     # contain upload_dir or /api/process will reject the uploaded paths.
367     upload_dir: str = "output/uploads"
368     # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
369     # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
370     max_upload_mb: int = 4096
371     max_part_mb: int = 95
372     allowed_upload_extensions: List[str] = ["mp4", "mov"]
373
374     # --- M9 edge auth (Cloudflare Access, D9) ---
375     # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
376     # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
377     # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
378     # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
379     # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
380     # application's audience tag. NO secrets here ? these are public identifiers.
381     edge_auth_enabled: bool = False
382     cf_team_domain: str = (
383         "" # e.g. https://<team>.cloudflareaccess.com (issuer + /certs JWKS)
384     )
385     cf_access_aud: str = "" # the CF Access application AUD tag the token must carry
386     # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
387     # backend/main.py so importing the app does no filesystem I/O; default ["*"].
Credentials stay
388     # OFF ? Cf-Access is a header, not a cookie.)
389
390
391 class StorageConfig(BaseModel):
392     """Pluggable storage backend (docs/DECOUPLED_COMPUTE/). Default ``local`` = today's
393     behaviour (filesystem paths, byte-for-byte). ``s3`` covers AWS + Cloudflare R2 + D0
Spaces
394     + MinIO (via ``endpoint_url``); ``gcs`` = Google Cloud Storage; ``gdrive`` = a
locally MOUNTED
395     Google Drive (``gdrive.mount_root``). Per-backend settings are loose dicts passed to
the constructor ?
396     **secrets are never stored here**, only endpoints / regions / file paths / source-
selectors
397     (boto3/google auth chains supply credentials). See ``backend/storage/``."""
398
399     backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
400     # Output root for produced artifacts. "" => fall back to output.output_dir (non-
breaking).
401     output_root: str = ""
402     # --- Input side (COMPUTE_DECOUPLED_SERVING M2; parallel to the output

```

```

`backend`/`output_root`).
403     # Default ``local`` / "" keeps today's behaviour: a bare path / ``local://`` URI is
read IN PLACE
404     # (identity, no copy). Set these to read inputs from a bucket / mounted Drive ? a
bare/relative
405     # input name is resolved against ``input_root`` (e.g.
``input_root``="gs://bucket/videos"`` + ".mp4"
406     # => ``gs://bucket/videos/x.mp4``) or, if ``input_root`` is empty, prefixed with
``input_backend``
407     # (``"gs"`` + "bucket/x.mp4" => ``gs://bucket/x.mp4``). An explicit scheme on the
input
408     # (``gs://?``/``s3://?``) or an absolute local path ALWAYS wins. Non-local inputs
are fetched to a
409     # scratch dir before processing (see ``backend.storage.acquire_local_input``).
410     input_backend: Literal["local", "s3", "gs", "gdrive", "gdrive-api"] = "local"
411     input_root: str = ""
412     # --- Per-video workspace (docs/PER_VIDEO_WORKSPACE.md) ---
413     # One folder per video keyed by the MD5 video_id:
<root>/[<workspace_prefix>]/<video_id>/...
414     # so a local deploy and a cloud (bucket) deploy share ONE relative layout (only the
root
415     # differs). DEFAULT-OFF: convergence is phased + flipped via a Launch Report (D5).
When OFF
416     # the legacy flat layout is used unchanged.
417     single_folder_per_video: bool = False
418     # Durable workspace root URI. "" => precedence: output_root, then output.output_dir.
419     workspace_root: str = ""
420     # Cloud namespace prefix under the root so per-video folders sit tidily beside
421     # videos/ labels/ weights/ (e.g. gs://bucket/workspaces/<video_id>/). "" = root
directly.
422     workspace_prefix: str = "workspaces"
423     local: dict[str, Any] = Field(default_factory=dict)
424     s3: dict[str, Any] = Field(default_factory=dict)
425     gcs: dict[str, Any] = Field(default_factory=dict)
426     gdrive: dict[str, Any] = Field(default_factory=dict)
427     # gdrive-api settings (e.g. {"root_folder_id": "<id>"}); read via the
hyphen?underscore lookup in
428     # backend.storage.get_storage. NO secrets ? creds come from ADC over the Drive
scope.
429     gdrive_api: dict[str, Any] = Field(default_factory=dict)
430
431
432     class DeploymentConfig(BaseModel):
433         """Deployment profile ? one switch that selects a coherent bundle of compute +
storage
434         knobs (COMPUTE_DECOUPLED_SERVING M1, ``docs/COMPUTE_DECOUPLED_SERVING/README.md``
$4.3).
435
436         **Pure / additive:** the empty default profile (``""``) applies NO preset, so the
pipeline
437         behaves exactly as before this section existed. A profile is opt-in (``config.json``
``deployment.profile`` or the ``DEPLOYMENT_PROFILE`` env var); auto-detect only ever
438         *suggests* one (D15 ? cloud spend is never entered silently). Nothing in the core
reads
439         these fields yet (M1): selecting a profile works purely by pre-setting the EXISTING
knobs
440         (``indexing.detector_impl`` / ``skip_ai_handoff``, ``storage.backend``) in the
loader.
441
442         Valid values mirror ``backend/config/presets.py`` (parity-pinned by a unit test)."""
443
444         # "" = no profile (today's manual knobs). Others apply presets.PROFILE_PRESETS.
445         profile: Literal["", "truly-local", "cloud-serving", "cpu-mock"] = ""
446         # Where the core runs. Set by the profile preset; overridable in config.json. Inert
in M1

```



```

448         # (recorded for observability; consumed by the ComputeBackend seam from M2+).
449         compute_target: Literal["", "local_gpu", "cloud_run", "cpu_mock"] = ""
450
451
452     class BillingConfig(BaseModel):
453         """Per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8, D8).
454
455         **DEFAULT-OFF:** ``enabled=False`` ? nothing records cost (the v6 cost columns stay
456         NULL = today's behaviour). When enabled, a job's cost is computed + stored in **USD** via
457         the versioned ``pricebook`` DB table and **displayed in INR** at ``fx_inr_per_usd``. The default
458         pricebook version is the seeded ``placeholder-v0`` (every rate ``0.0`` ? cost ``0``) until the
459         owner inserts a real, calibrated version and points ``pricebook_version`` at it.
460
461         ? Even with ``enabled=True`` + a real pricebook, recorded cost stays **0** until a
462         follow-up wires the worker to emit per-job ``usage`` (gpu/wall/egress/gemini) ? M8 lands the
463         ledger + pricebook + the recording seam; usage emission (the metering hooks) is a separate
464         step."""
465         enabled: bool = False
466         pricebook_version: str = (
467             "placeholder-v0" # the seeded $0 version; owner flips to a real one
468         )
469         fx_inr_per_usd: float = 83.0 # USD?INR display rate (Bangalore market)
470         margin: float = 0.0 # resale markup (?0); 0 = bill at cost
471
472
473     class FlywheelConfig(BaseModel):
474         """Data-flywheel harness (COMPUTE_DECOUPLED_SERVING M11, D12) ? the
475         consent?capture?eval?goldenize
476         loop: on a CONSENTED adult job, durably **capture** the source + rally output into
477         the per-video workspace ? **shadow-eval** owned-vs-Gemini ? **propose promotion** to the golden
478         TRAINING set, so the owned model climbs toward the D11 ship floor.
479
480         **DEFAULT-OFF + INERT:** ``enabled=False`` ? nothing is captured/evaluated (byte-
481         identical to today). ? **Consent is FAKED-DENY until counsel + the consent schema land:** even
482         ``enabled=True`` captures **nothing**, because ``flywheel.consent_attested`` currently returns
483         ``False`` for every job (the privacy guardrail ? D10: adults-only, explicit opt-in, derived-data + auto-
484         delete-raw). This is the PLUMBING only; the real consent gate (counsel-reviewed ToS/DPA ? consent
485         columns ? wiring) + the owned-model + golden-label data needed to *activate* shadow-
486         eval/promotion are owner/counsel/GPU scope. Pairs with M9-consent."""
487         enabled: bool = False
488         workspace_prefix: str = (
489             "flywheel" # per-video workspace prefix for captured artifacts
490         )
491         shadow_eval: bool = True # run the owned-vs-Gemini shadow comparison on capture
492         min_promote_n: int = 5 # promote_if_served_safe `min_n` (the served-safe gate)
493
494
495     class AppConfig(BaseModel):
496         """Root configuration object.

```

```

497
498     All runtime configuration should be accessed via an instance of this class
499     (or the loader) rather than raw dict lookups.
500     """
501
502     sport: Sport = "badminton"
503     ai_provider: str = "gemini"
504     # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
flagship;
505     # the alias never deprecates underneath us. Eval reports record the run date
alongside.
506     ai_model: str = "gemini-flash-latest"
507
508     validation: ValidationConfig = Field(default_factory=ValidationConfig)
509     indexing: IndexingConfig = Field(default_factory=IndexingConfig)
510     output: OutputConfig = Field(default_factory=OutputConfig)
511     stitching: StitchingConfig = Field(default_factory=StitchingConfig)
512     security: SecurityConfig = Field(default_factory=SecurityConfig)
513     logging: LoggingConfig = Field(default_factory=LoggingConfig)
514     storage: StorageConfig = Field(default_factory=StorageConfig)
515     deployment: DeploymentConfig = Field(default_factory=DeploymentConfig)
516     billing: BillingConfig = Field(default_factory=BillingConfig)
517     flywheel: FlywheelConfig = Field(default_factory=FlywheelConfig)
518
519     # Reject unknown keys ? typed config migration is complete (D3).
520     model_config = {
521         "extra": "forbid",
522         "validate_assignment": True,
523     }
524
525
526     # Convenience type alias for callers that want to annotate
527     Config = AppConfig
528

```

9. user (2026-07-04T02:26:14.302Z)

```

1     """Startup checks ? per-profile coherence (M5) + exposure-safety posture (M9 hardening).
2
3     A deployment profile (`truly-local` / `cloud-serving` / `cpu-mock`) implies a
coherent
4     runtime. These checks fail/warn fast at process startup ? a loud signal before any
5     (expensive) job runs ? when the active profile is inconsistent with the environment:
6
7     * cloud-serving with a non-cloud storage backend ? a hard error (the one
config bug we
8     can detect with certainty + zero false-positive risk: a cloud profile that can't
reach a
9     bucket can never serve). This is the README §4.3 "loud error before any job".
10    * a GPU mismatch (e.g. `device='cuda'` on a box where CUDA isn't visible, or
`device='auto'`
11    falling back to CPU) ? a warning, never an error: the probe is best-effort
(torch may live
12    only in the detector subprocess env ? see `presets.probe_cuda_available`), and the
detector's
13    own healthcheck is the real authority. A warning surfaces it early without false-
failing.
14    * creds that can't be confirmed (no `GOOGLE_APPLICATION_CREDENTIALS` / not on
Cloud Run) ?
15    a warning: Application Default Credentials can come from the metadata server /
gcloud, which
16    we can't probe offline, so the storage backend's own init is the authority; we just
flag it.
17
18    Exposure-safety posture (server only, `include_exposure=True`). When the engine is

```

```

*exposed*
19     behind an edge gate (the owner flips ``security.edge_auth_enabled`` on for the
Cloudflare-Access
20     boundary ? the alpha-shareable "tunnel-to-a-box" topology), the gate is only half the
safety story.
21     A half-configured exposure starts *looking* healthy then leaks / 500s per request, so we
verify the
22     rest of the posture at **boot**: the ``allowed_video_root`` path-jail is set (else an
authenticated
23     tenant could read an arbitrary local file ? **error**), the CF team-domain + AUD are
present (else
24     no token can verify ? **error**, lifting the current first-request failure to boot), and
the
25     ``auth`` extra is importable (else every request 500s ? **warning**). Only the HTTP
**server** passes
26     ``include_exposure=True``; the worker reads ``--video-uri`` (not untrusted tenant
paths), so it is
27     never "exposed" and stays unaffected.
28
29     **DEFAULT-OFF (both dimensions):** with no profile selected (``profile == ""``) the
profile checks are
30     a strict **no-op** ? exactly the pre-M5 behaviour; with ``edge_auth_enabled=False``
(today's default)
31     the exposure checks are a strict **no-op** too. So nothing changes for today's manual-
knob deployments.
32     ""
33
34     from __future__ import annotations
35
36     import logging
37     import os
38     from dataclasses import dataclass
39     from typing import Any, Mapping, Optional
40
41     LEVEL_ERROR = "error"
42     LEVEL_WARN = "warn"
43
44     # Storage backends that count as "a cloud bucket" for cloud-serving coherence.
45     _CLOUD_STORAGE = ("gcs", "s3")
46     # STRONG env signals that Google ADC *might* resolve. Deliberately NOT
GOOGLE_CLOUD_PROJECT /
47     # CLOUDSDK_CONFIG ? those are often set without usable creds (a plain project id / a
config dir
48     # with no active login), so they would suppress the heads-up falsely. (s3 creds are
intentionally
49     # left to boto3's default chain ? no parallel heads-up, by design.)
50     _GCP_ENV_HINTS = ("GOOGLE_APPLICATION_CREDENTIALS", "K_SERVICE", "GCE_METADATA_HOST")
51
52
53     @dataclass(frozen=True)
54     class StartupCheck:
55         """One per-profile finding. ``level`` is ``error`` (fatal) or ``warn`` (heads-
up)."""
56
57         level: str
58         code: str # stable, greppable code (e.g. "STORAGE_INCOHERENT")
59         message: str
60
61
62     class StartupCheckError(RuntimeError):
63         """Raised by ``enforce_startup_checks`` when an active profile fails a FATAL
check."""
64
65     def __init__(self, failures: list[StartupCheck]):
66         self.failures = failures

```

```

67         super().__init__(
68             "deployment profile startup check failed: "
69             + "; ".join(f"[{c.code}] {c.message}" for c in failures)
70         )
71
72
73     def _creds_discoverable(env: Mapping[str, str]) -> bool:
74         """True if *some* Google ADC source is hinted by the environment. Best-effort ? a
False here
75         only downgrades to a WARNING, never a hard failure (the storage init is the
authority)."""
76         return any(env.get(k) for k in _GCP_ENV_HINTS)
77
78
79     def _auth_extra_available() -> bool:
80         """True if PyJWT (the ``auth`` extra) is importable ? probed WITHOUT importing it
(find_spec),
81         so this stays cheap + side-effect-free. Module-level so tests can monkeypatch it
deterministically
82         (whether PyJWT is installed in the test env must not change the WARN's presence)."""
83         try:
84             import importlib.util
85
86             return importlib.util.find_spec("jwt") is not None
87         except (
88             Exception
89         ): # pragma: no cover - find_spec only raises on a broken parent package
90             return False
91
92
93     def _exposure_checks(config: Any, env: Mapping[str, str]) -> list[StartupCheck]:
94         """Exposure-safety posture for an *exposed* HTTP server. Gated on
``security.edge_auth_enabled``
95         (default False ? ``[]`` = strict no-op, today's behaviour). When edge auth is ON ?
the "I am
96         exposing this engine behind the CF-Access gate" signal ? the gate alone isn't
enough: verify the
97         rest of the posture at boot so a half-configured exposure fails fast instead of
leaking / 500-ing
98         per request. Config-only (no IO beyond the ``find_spec`` auth-extra probe); ``env``
is the
99         caller's environment view (so the host-allowlist check is testable)."""
100         sec = getattr(config, "security", None)
101         if not sec or not getattr(sec, "edge_auth_enabled", False):
102             return [] # DEFAULT-OFF: edge auth off ? no posture enforcement (byte-identical
to today).
103
104         out: list[StartupCheck] = []
105
106         # (1) The path-jail is load-bearing once authenticated tenants can hit /api/process:
WITHOUT it,
107         # a tenant can point the engine at an ARBITRARY local file (allowed_video_root unset
= "any local
108         # file", the single-user default). Fatal for an exposed engine.
109         root = getattr(sec, "allowed_video_root", None)
110         if not root or not str(root).strip():
111             out.append(
112                 StartupCheck(
113                     LEVEL_ERROR,
114                     "EXPOSED_NO_VIDEO_ROOT",
115                     "security.edge_auth_enabled is ON (engine exposed) but
security.allowed_video_root is "
116                     "unset ? an authenticated tenant could make /api/process read an
arbitrary local file. "
117                     "Set allowed_video_root to a jail directory before exposing the

```

```

engine.",
118         )
119     )
120
121     # (2) Edge auth cannot verify a token without the team domain + AUD. Today that only
fails on the
122     # FIRST request (auth.resolve_tenant); lift it to BOOT so a misconfigured gate never
starts
123     # "healthy" then 401s every request.
124     team = (getattr(sec, "cf_team_domain", "") or "").strip()
125     aud = (getattr(sec, "cf_access_aud", "") or "").strip()
126     if not team or not aud:
127         missing = " + ".join(
128             n for n, v in (("cf_team_domain", team), ("cf_access_aud", aud)) if not v
129         )
130     out.append(
131         StartupCheck(
132             LEVEL_ERROR,
133             "EXPOSED_EDGE_AUTH_INCOMPLETE",
134             f"security.edge_auth_enabled is ON but {missing} is not configured ? the
Cf-Access JWT "
135             "cannot be verified (every request would fail). Set
security.cf_team_domain "
136             "(e.g. https://<team>.cloudflareaccess.com) and security.cf_access_aud
(the CF Access "
137             "application AUD tag).",
138         )
139     )
140
141     # (3) The auth extra (PyJWT[crypto]) is lazy-imported by
auth.verify_cf_access_token. If it's
142     # missing, every request 500s at verify time ? a boot WARN beats a per-request 500.
Warning, not
143     # fatal: the auth module's own ImportError is the final authority.
144     if not _auth_extra_available():
145         out.append(
146             StartupCheck(
147                 LEVEL_WARN,
148                 "EXPOSED_AUTH_EXTRA_MISSING",
149                 "security.edge_auth_enabled is ON but PyJWT[crypto] is not importable ?
Cf-Access "
150                 "verification will fail at request time. Install the 'auth' extra (pip
install -e .[auth]).",
151             )
152         )
153
154     # (4) P0c: an exposed engine binds 0.0.0.0, but the Host-header (DNS-rebinding)
guard defaults to
155     # loopback-only when RALLY_ALLOWED_HOSTS is unset (see main._allowed_hosts_from_env)
? it would
156     # then 400 every request the edge forwards with its public Host. Boot WARN beats a
silent 400-all.
157     if not (env.get("RALLY_ALLOWED_HOSTS", "") or "").strip():
158         out.append(
159             StartupCheck(
160                 LEVEL_WARN,
161                 "EXPOSED_HOST_ALLOWLIST_LOOPBACK",
162                 "security.edge_auth_enabled is ON but RALLY_ALLOWED_HOSTS is unset ? the
Host-header guard "
163                 "defaults to loopback-only and will 400 forwarded requests. Set
RALLY_ALLOWED_HOSTS to your "
164                 "public host(s) (or '*' when the edge gate validates the host).",
165             )
166         )
167

```

```

168         return out
169
170
171     def run_startup_checks(
172         config: Any,
173         *,
174         cuda_available: Optional[bool] = None,
175         env: Optional[Mapping[str, str]] = None,
176         include_exposure: bool = False,
177     ) -> list[StartupCheck]:
178         """Return the startup findings for ``config``.
179
180         ``cuda_available`` is the caller's best-effort GPU probe (``None`` = not probed ?
GPU checks
181         are skipped; the API server passes ``None``, the worker passes
``presets.probe_cuda_available()``).
182         ``include_exposure`` adds the exposure-safety posture (``_exposure_checks``) ? the
HTTP **server**
183         passes ``True``; the worker leaves it ``False`` (it is never "exposed"). Both
dimensions are
184         DEFAULT-OFF: no profile ? no profile checks; ``edge_auth_enabled=False`` ? no
exposure checks.
185         Pure: no IO (beyond the ``find_spec`` auth probe), no torch import ? the caller owns
the GPU probe.
186         """
187         e = os.environ if env is None else env
188         checks: list[StartupCheck] = []
189
190         # Exposure-safety posture ? INDEPENDENT of deployment.profile (a tunnel-to-a-box
exposure often
191         # runs with no profile set), gated on edge_auth_enabled (default-OFF). Server-only.
192         if include_exposure:
193             checks.extend(_exposure_checks(config, e))
194
195         profile = getattr(getattr(config, "deployment", None), "profile", "") or ""
196         if not profile:
197             return checks # no profile ? only the exposure posture (itself off unless edge
auth is on).
198
199         storage_backend = getattr(getattr(config, "storage", None), "backend", "local")
200         indexing = getattr(config, "indexing", None)
201         detector_impl = getattr(indexing, "detector_impl", "auto")
202         device = getattr(getattr(indexing, "wasb", None), "device", "cuda")
203
204         if profile == "truly-local":
205             # truly-local reads bytes from the local FS or a *mounted* Drive ? a cloud
backend here is
206             # an inconsistent (but not fatal) choice worth surfacing.
207             if storage_backend not in ("local", "gdrive"):
208                 checks.append(
209                     StartupCheck(
210                         LEVEL_WARN,
211                         "STORAGE_UNUSUAL",
212                         f"profile 'truly-local' usually uses local/gdrive storage, but
storage.backend="
213                         f"'{storage_backend}'. Reads will go to a remote backend.",
214                     )
215                 )
216             # GPU heads-up (warning only ? the detector healthcheck is the authority; the
probe is
217             # best-effort and may be False even when a subprocess detector env has CUDA).
218             if cuda_available is False:
219                 if device == "cuda":
220                     checks.append(
221                         StartupCheck(

```

```

222             LEVEL_WARN,
223             "CUDA_NOT_VISIBLE",
224             "device='cuda' but no CUDA GPU is visible to this process. If
the box truly "
225             "has no GPU the detector healthcheck will hard-fail ? set
indexing.wasb.device="
226             "'auto' for the CPU fallback (M4).",
227         )
228     )
229     elif device == "auto":
230         checks.append(
231             StartupCheck(
232                 LEVEL_WARN,
233                 "CPU_FALLBACK",
234                 "device='auto' and no CUDA GPU is visible to THIS process ? auto
may resolve "
235                 "to CPU (much slower). If torch lives only in the detector
subprocess env this "
236                 "can be a false alarm; the runner's DeviceContext/healthcheck is
the authority.",
237             )
238         )
239
240     elif profile == "cloud-serving":
241         # The ONE fatal coherence check: a cloud profile that can't reach a bucket can't
serve.
242         if storage_backend not in _CLOUD_STORAGE:
243             checks.append(
244                 StartupCheck(
245                     LEVEL_ERROR,
246                     "STORAGE_INCOHERENT",
247                     f"profile 'cloud-serving' requires a cloud storage backend "
248                     f"('{','.join(_CLOUD_STORAGE)}'), but
storage.backend='{storage_backend}'. "
249                     "A cloud job cannot read/write a local-only backend.",
250                 )
251             )
252         elif storage_backend == "gcs" and not _creds_discoverable(e):
253             checks.append(
254                 StartupCheck(
255                     LEVEL_WARN,
256                     "CREDS_UNCONFIRMED",
257                     "no GOOGLE_APPLICATION_CREDENTIALS / Cloud Run env detected; relying
on Application "
258                     "Default Credentials. The GCS backend will fail loudly at init if
ADC can't resolve.",
259                 )
260             )
261             # The AI-handoff key: with skip_ai_handoff=False the segmenter demands
262             # ${AI_PROVIDER}_API_KEY at rally time and bails to an EMPTY result when it is
263             # missing ? a silent 0-rally "success" AFTER the GPU pass is paid for (observed
on
264             # the first real cloud-serving litmus, 2026-07-03). The env cannot appear later
in
265             # a running container, so this is a zero-false-positive boot-time fatal.
266             skip_ai = bool(getattr(indexing, "skip_ai_handoff", False))
267             provider = str(getattr(config, "ai_provider", "gemini") or "gemini")
268             key_var = f"{provider.upper()}_API_KEY"
269             if not skip_ai and not e.get(key_var):
270                 checks.append(
271                     StartupCheck(
272                         LEVEL_ERROR,
273                         "AI_HANDOFF_KEY_MISSING",
274                         f"profile 'cloud-serving' runs the {provider} handoff "
275                         f"({indexing.skip_ai_handoff=False}) but {key_var} is not set ? the

```

```

segmenter "
276             f"would bail to an empty result after the paid GPU pass. Mount the
key "
277             f"(e.g. a Secret Manager secret) or set
indexing.skip_ai_handoff=true for "
278             "the fully-local candidates-as-rallies mode.",
279         )
280     )
281     if device == "cpu":
282         checks.append(
283             StartupCheck(
284                 LEVEL_WARN,
285                 "CLOUD_CPU_DEVICE",
286                 "profile 'cloud-serving' runs on a GPU (L4) but
indexing.wasb.device='cpu' ? "
287                 "detection will not use the GPU.",
288             )
289         )
290     elif cuda_available is False and device in ("cuda", "auto"):
291         checks.append(
292             StartupCheck(
293                 LEVEL_WARN,
294                 "CLOUD_CUDA_NOT_VISIBLE",
295                 "profile 'cloud-serving' expects a GPU but no CUDA GPU is visible to
this process.",
296             )
297         )
298
299     elif profile == "cpu-mock":
300         # The mock profile is for CI/regression ? a real detector here defeats the
purpose.
301         if detector_impl != "stub":
302             checks.append(
303                 StartupCheck(
304                     LEVEL_WARN,
305                     "MOCK_NONSTUB_DETECTOR",
306                     f"profile 'cpu-mock' normally uses detector_impl='stub', but
detector_impl="
307                     f"'{detector_impl}'. A real detector defeats the GPU-free mock.",
308                 )
309             )
310
311     return checks
312
313
314     def enforce_startup_checks(
315         config: Any,
316         *,
317         cuda_available: Optional[bool] = None,
318         env: Optional[Mapping[str, str]] = None,
319         raise_on_error: bool = True,
320         include_exposure: bool = False,
321         logger: Optional[logging.Logger] = None,
322     ) -> list[StartupCheck]:
323         """Run the checks, LOG each (errors at ERROR, warnings at WARNING), and ? unless
324         ``raise_on_error=False`` ? raise ``StartupCheckError`` if any FATAL check tripped.
325
326         ``include_exposure`` (server-only) adds the exposure-safety posture. Returns the
full findings
327         list (so a caller can inspect/surface them). A no-op with no logging and no raise
when both
328         dimensions are off (``run_startup_checks`` returns ``[]``)."""
329         log = logger or logging.getLogger(__name__)
330         checks = run_startup_checks(
331             config,

```



```

332         cuda_available=cuda_available,
333         env=env,
334         include_exposure=include_exposure,
335     )
336     for c in checks:
337         if c.level == LEVEL_ERROR:
338             log.error("[STARTUP] %s: %s", c.code, c.message)
339         else:
340             log.warning("[STARTUP] %s: %s", c.code, c.message)
341     errors = [c for c in checks if c.level == LEVEL_ERROR]
342     if errors and raise_on_error:
343         raise StartupCheckError(errors)
344     return checks
345

```

10. user (2026-07-04T02:26:14.779Z)

```

1     """Server / network bootstrap helpers, extracted from ``backend.main`` (P24 god-module
split).
2
3     Pure, side-effect-free helpers for the CORS/host-allowlist middleware config and the
friendly
4     ``indexer --app`` launcher (bind-host precedence, free-port fallback, open-browser-when-
ready).
5     No module globals, no ``app`` object, no ``backend.main`` import ? a plain move;
``backend.main``
6     re-imports these names (ABOVE its ``app = FastAPI(...)`` so the module-level
``add_middleware``
7     calls still resolve them) and re-exports them so ``import backend.main as m;
m._resolve_bind_host``
8     and the like keep working unchanged.
9     """
10
11     import os
12     import socket
13     import webbrowser
14     from typing import Dict, List, Optional
15
16     from backend.config import AppConfig
17
18
19     def _cors_origins_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
20         """M9: CORS allow-origins from the ``RALLY_CORS_ALLOW_ORIGINS`` env var (comma-
separated),
21         default ``["*"]``. Read from the ENV (not a cwd ``config.json``) so importing
``backend.main``
22         does zero filesystem I/O ? a hosted/multi-tenant deploy sets this to lock CORS to
its origin."""
23         src = os.environ if env is None else env
24         raw = (src.get("RALLY_CORS_ALLOW_ORIGINS", "") or "").strip()
25         origins = [o.strip() for o in raw.split(",") if o.strip()]
26         return origins or ["*"]
27
28
29     def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
30         """P0c: Host-header allowlist ? a DNS-rebinding guard for the localhost appliance. A
malicious
31         web page can point a domain it controls at 127.0.0.1 and script requests to the
local server;
32         the browser sends ``Host: attacker.com``, which this rejects (the server only
answers loopback
33         Host headers by default). ``RALLY_ALLOWED_HOSTS`` (comma-separated) overrides; the
default keeps
34         Starlette's TestClient host ``testserver`` so the in-process suite isn't broken.
Read from the

```

```

35     ENV (not config.json) so importing ``backend.main`` does zero filesystem I/O; a
routable/edge
36     deploy sets its public host (or ``*`` when fronted by an auth gate that validates
the host)."""
37     src = os.environ if env is None else env
38     raw = (src.get("RALLY_ALLOWED_HOSTS", "") or "").strip()
39     hosts = [h.strip() for h in raw.split(",") if h.strip()]
40     # IPv4 loopback (the default bind) + the TestClient host. NB: Starlette's
TrustedHostMiddleware
41     # matches on host.split(":")[0], which mangles a bracketed IPv6 Host ("[::1]:port"),
so IPv6
42     # loopback is NOT supported by this guard ? access the appliance via
127.0.0.1/localhost.
43     return hosts or ["localhost", "127.0.0.1", "testserver"]
44
45
46     def _resolve_bind_host(
47         config: Optional[AppConfig] = None, cli_host: Optional[str] = None
48     ) -> str:
49         """P0c: the server bind address. Precedence: ``--host`` > ``RALLY_BIND_HOST`` env >
``0.0.0.0``
50         when an edge-auth profile is active (``security.edge_auth_enabled`` ? i.e. there IS
an auth gate
51         in front) > ``127.0.0.1`` (default: localhost-only, today's behaviour). Binding a
routable
52         interface with no auth gate is unsafe, hence the localhost default; widen it
consciously."""
53         if cli_host and cli_host.strip():
54             return cli_host.strip()
55         env = os.environ.get("RALLY_BIND_HOST")
56         if env and env.strip():
57             return env.strip()
58         sec = getattr(config, "security", None)
59         if getattr(sec, "edge_auth_enabled", False):
60             return "0.0.0.0" # behind a Cloudflare-Access-style edge gate (M9) ? bind all
interfaces
61         return "127.0.0.1"
62
63
64     def _port_is_free(host: str, port: int) -> bool:
65         """True if (host, port) can be bound right now (best-effort; a TOCTOU race is
acceptable ?
66         uvicorn surfaces a real bind failure loudly)."""
67         with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
68             try:
69                 s.bind((host, port))
70                 return True
71             except OSError:
72                 return False
73
74
75     def _find_free_port(host: str) -> int:
76         """An OS-assigned free port on host."""
77         with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
78             s.bind((host, 0))
79             return int(s.getsockname()[1])
80
81
82     def _resolve_app_port(host: str, port: int) -> int:
83         """For --app: keep `port` if free, else fall back to a free one so a busy 8000 never
blocks the app."""
84         return port if _port_is_free(host, port) else _find_free_port(host)
85
86
87     def _open_browser_when_ready(url: str, health_url: str, timeout: float = 60.0) -> bool:

```

```

88         """Poll `health_url` until the server is up, then open `url` in the user's default
browser.
89         Returns True if a browser open was attempted. No-op (returns False) when
RALLY_NO_BROWSER is set
90         (tests / headless) or the server never becomes healthy (don't open a dead page).
Never raises."""
91         if os.environ.get("RALLY_NO_BROWSER"):
92             return False
93         import time as _time
94         import urllib.request
95
96         deadline = _time.monotonic() + timeout
97         while _time.monotonic() < deadline:
98             try:
99                 with urllib.request.urlopen(health_url, timeout=2) as r:
100                     if getattr(r, "status", 200) == 200:
101                         break
102             except Exception:
103                 _time.sleep(0.3)
104         else:
105             return False
106         try:
107             webbrowser.open(url)
108         except Exception:
109             pass
110         return True
111

```

11. assistant (2026-07-04T02:26:16.606Z)

Now let me check the Dockerfile and relevant docs:

12. user (2026-07-04T02:26:17.270Z)

```

1      # Cloud Run GPU **Job** image for the rally worker (COMPUTE_DECOUPLED_SERVING M6/M7,
ADR-012/D16).
2      #
3      # Runs `python -m backend.worker infer ?` INLINE on an L4 (detect + window + stitch co-
located so
4      # CPU/wall + egress are metered into one job ? D4). Built + pushed + verified by the
owner on the
5      # M7 real run (the $100/?10k budget, D17); CI does NOT build this (a CUDA image is too
heavy) ? the
6      # dispatch shim is proven with a fake client (tests/test_compute_backend.py).
7      #
8      # Two python stacks, deliberately separate (a framework conflict otherwise ? same split
as the
9      # native GPU box, deploy/digitalocean/gpu_setup.sh):
10     # * the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching;
11     # * a conda `wasb` env (py3.8 + torch 1.11.0+cu113) runs the WASB shuttle detector as
a subprocess
12     # (the native runner shells out to $WASB_PYTHON), so the app's torch never co-
installs with it.
13     #
14     # ? WASB-C3: the pretrained badminton weights are academic-data-trained ? provenance NOT
cleared for
15     # commercial sharing. They are NOT baked into this image; stage them at runtime (a
bucket fetch or a
16     # mount) and point $WASB_WEIGHTS at them. Resolve WASB-C3 before any public, non-owner
serving.
17
18     FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04
19
20     ENV DEBIAN_FRONTEND=noninteractive \
21         PYTHONUNBUFFERED=1 \

```

```

22     PIP_NO_CACHE_DIR=1 \
23     CONDA_DIR=/opt/conda \
24     WASB_REPO_DIR=/opt/models/WASB-SBDT
25
26     # ffmpeg (stitch/decode) + DejaVu fonts (the watermark filter) + git/curl for the WASB
env build.
27     RUN apt-get update && apt-get install -y --no-install-recommends \
28         ffmpeg fonts-dejavu-core git curl ca-certificates build-essential \
29         python3 python3-pip python3-venv \
30         && rm -rf /var/lib/apt/lists/*
31
32     # --- WASB detector env (micromamba + conda-forge ? BSD-licensed, NO Anaconda ToS; P3a,
ADR-005) ---
33     # Enforces ADR-005 (every dependency ? code, data, weights, AND the build toolchain ?
stays
34     # MIT/Apache/BSD): micromamba (BSD-3) + conda-forge (community) replaces Miniconda's
Anaconda-ToS-gated
35     # `defaults` channels (the ?200-employee Terms-of-Service trap flagged in
PACKAGING_PLAN.md §3).
36     # MAMBA_ROOT_PREFIX=$CONDA_DIR (= /opt/conda) keeps the env at $CONDA_DIR/envs/wasb so
$WASB_PYTHON below
37     # is UNCHANGED. Same verified torch-1.11.0+cu113 WASB stack; the pip wheels are
unaffected by the switch.
38     ENV MAMBA_ROOT_PREFIX=/opt/conda
39     RUN curl -sSL https://micro.mamba.pm/api/micromamba/linux-64/latest \
40         | tar -xj -C /usr/local bin/micromamba \
41         && micromamba create -y -q -n wasb -c conda-forge python=3.8 \
42         && git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO_DIR" \
43         && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
44             torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113 \
45         && ( [ -f "$WASB_REPO_DIR/requirements.txt" ] && "$CONDA_DIR/envs/wasb/bin/python"
-m pip install -q -r "$WASB_REPO_DIR/requirements.txt" || true ) \
46         && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
47             hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml
matplotlib "numpy<1.24"
48
49     # --- Main application env ---
50     WORKDIR /app
51     COPY pyproject.toml README.md ./
52     COPY requirements-trackers.txt ./
53     COPY backend ./backend
54     COPY training ./training
55     # Install with the [storage-gcs] extra: the cloud worker PULLS the gs:// input and
PUSHES the
56     # gs:// outputs/done-marker through backend.storage (google-cloud-storage is lazy-
imported there).
57     # Without it the job dies at the first gs:// fetch with `ModuleNotFoundError:
google.cloud` ? the
58     # real-Cloud-Run failure the mock-client tests can't see (the bucket I/O is faked in
tests).
59     # D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ?
60     # roboflow/trackers#474) so it's NOT pulled by `pip install .`; install it out-of-band
before
61     # the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable
git SHA is
62     # single-sourced in requirements-trackers.txt.
63     RUN python3 -m pip install --upgrade pip \
64         && python3 -m pip install -r requirements-trackers.txt \
65         && python3 -m pip install ".[storage-gcs]"
66
67     # The native runner finds the WASB env + repo by env (no `conda activate` needed);
weights come in at
68     # runtime (WASB-C3). detector_impl=native = the Linux-native WASB runner; cuda stays the
default.

```

```

69     # PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until
process
70     # exit, so Cloud Logging shows a single end-of-run burst instead of live progress
(observed on the
71     # first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log
nothing.
72     ENV PYTHONUNBUFFERED=1 \
73         WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
74         WASB_REPO_DIR=/opt/models/WASB-SBDT \
75         WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
76         RALLY_DETECTOR_IMPL=native
77
78     # The dispatcher (CloudRunCompute) supplies the per-execution args:
79     # infer --video-uri gs:/// --out-uri gs:/// --done-uri gs:/// --segmenter wasb_hybrid
--set ?
80     ENTRYPOINT ["python3", "-m", "backend.worker"]
81

```

13. user (2026-07-04T02:26:17.750Z)

```

1     # Cloud Run GPU serving ? build / deploy / run (COMPUTE_DECOUPLED_SERVING M6/M7)
2
3     The cloud-serving compute path: a control plane (`CloudRunCompute`,
`backend/compute/cloud_run.py`)
4     dispatches a Cloud Run Job execution running this image's worker on an L4, then
bucket-polls
5     a done-marker for completion (D16). Scale-to-zero (D7) ? no warm pool.
6
7     > Status: the dispatch shim + image are landed + mock-tested (M6). The real
build + push +
8     > run on an L4 is the M7 owner step, within the $100/?10k budget (D17) ?
confirm cmd + hourly
9     > cost first, delete every resource after ([[gcp-vm-always-delete]]). CI does
not build this image.
10
11     ## 0. Prerequisites (owner)
12     - A GCP project with billing on + the Cloud Run, Artifact Registry, and Cloud Build APIs
enabled.
13     - GPU quota for L4 (`GPUS_ALL_REGIONS` ? 1) in the chosen region (asia-south1, else
Singapore ? see
14     `deploy/gcp/README.md`; L4 is often stocked-out, fall back per that runbook).
15     - The WASB weights staged in the bucket (WASB-C3 unresolved ? owner-gated):
`gs://<bucket>/models/wasb_badminton_best.pth.tar`.
16
17     ## 1. Build + push the image (Artifact Registry)
18     ```bash
19     PROJECT=<gcp-project>; REPO=rally
20     # L4 GPU is region-limited ? use asia-southeast1 (Singapore), NOT asia-south1 (Mumbai).
21     REGION=asia-southeast1
22     gcloud artifacts repositories create $REPO --repository-format=docker --location=$REGION
2>/dev/null || true
23     # `gcloud builds submit` has NO -f flag, so a non-root Dockerfile needs a build config:
24     gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project $PROJECT .
25     ```
26     *(A CUDA + micromamba image is large; the first build is ~8?10 min. Budget-watch per
D17.)*
27
28     ## 2. Create the Cloud Run Job (GPU, scale-to-zero)
29     The proven shape (first real M7 deploy, 2026-07-03): weights staged in a same-region
bucket
30     mounted read-only via GCS FUSE (no weights in the image ? WASB-C3; no init-fetch step
needed).
31     ```bash
32     gcloud run jobs create rally-worker \
33         --image $REGION-docker.pkg.dev/$PROJECT/$REPO/$IMG:latest \

```

```

34     --region $REGION --gpu 1 --gpu-type nvidia-l4 --no-gpu-zonal-redundancy \
35     --cpu 8 --memory 32Gi \
36     --max-retries 0 --task-timeout 3600 \
37     --set-env-vars WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar \
38     --add-volume name=serving,type=cloud-storage,bucket=$SERVING_BUCKET \
39     --add-volume-mount volume=serving,mount-path=/gcs
40     ``
41     - **`--no-gpu-zonal-redundancy` is REQUIRED on the CLI**, not just a cost choice (it
also halves
42     the GPU rate ? right for batch jobs): without the flag `gcloud` asks an interactive
Y/n which,
43     in a non-interactive shell, **hangs forever with zero output and nothing created**. An
empty
44     `gcloud` that "did nothing" ? suspect a hidden prompt.
45     - 8 vCPU / 32 GiB: decode + x264 are CPU-side (NVDEC is a deferred D16 knob), so CPU
headroom is
46     what keeps the paid GPU busy. `/tmp` is **RAM-backed** ? the pulled input video lives
in memory,
47     so size `--memory` for the largest expected upload (a 5.8 GB GoPro file ? 6 GiB of
tmpfs).
48
49     ## 3. Wire the dispatcher (control plane)
50     The control plane runs with the **cloud-serving** profile (`DEPLOYMENT_PROFILE=cloud-
serving` ?
51     `compute_target=cloud_run`). `get_compute_backend(config)` then builds `CloudRunCompute`
from env:
52     ```bash
53     export DEPLOYMENT_PROFILE=cloud-serving
54     export CLOUD_RUN_JOB=rally-worker CLOUD_RUN_REGION=$REGION
GOOGLE_CLOUD_PROJECT=$PROJECT
55     # A `worker infer` (or the API job path) with a gs:// video now DISPATCHES to the Job
instead of
56     # running in-process; the container runs the SAME worker INLINE (compute_target forced
local_gpu).
57     python -m backend.worker infer --video-uri gs://<bucket>/videos/clip.mp4 --out-uri
gs://<bucket> --segmenter wasb_hybrid
58     ``
59     **The AI handoff needs its key IN THE JOB env** (`GEMINI_API_KEY` ? mount a Secret
Manager secret
60     on the job), or run fully-local with `--set indexing.skip_ai_handoff=true` (candidate
windows
61     emitted as rallies ? the eval-parity mode). With neither, the segmenter used to bail to
a silent
62     0-rally "success" **after** the paid GPU pass; the cloud-serving startup checks now fail
this at
63     boot (`AI_HANDOFF_KEY_MISSING`, rc 2) before the video is even pulled.
64
65     ### 3a. Control-plane SERVICE deploy ? required flags for reliable async jobs
66
67     The API server (`/api/process`, `/api/compile` ? #329) runs its **job worker IN-
PROCESS**: the
68     request returns `202` and a background thread drains the job (GPU detection dispatches
to the Job;
69     the reel stitch runs on the control plane itself). On Cloud Run that background thread
only makes
70     progress while the instance has CPU, so a scale-to-zero **service** deploy MUST set:
71     ```bash
72     gcloud run deploy rally-control-plane --image <image> --region $REGION \
73     --no-cpu-throttling \           # CPU stays allocated between requests ? the drain thread
finishes
74     --min-instances 1 \           # one always-warm instance drains reliably (cheap CPU;
NOT the D7 GPU pool)
75     --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
76     --set-env-vars DEPLOYMENT_PROFILE=cloud-serving,CLOUD_RUN_JOB=rally-
worker,CLOUD_RUN_REGION=$REGION,GOOGLE_CLOUD_PROJECT=$PROJECT

```

```

77     ``
78     Without `--no-cpu-throttling`, a `/api/compile` returns 202 and then **stalls** ? the
reel never
79     finishes and startup crash-recovery would mark it terminal. `min-instances 1` does NOT
contradict
80     D7 (that was the ~$500/mo warm **GPU** pool); this is a ~1-vCPU CPU instance. Defense-
in-depth is in
81     the code: `Database.recover_stale_jobs()` **re-queues** a compile interrupted by a
restart (idempotent
82     stitch) rather than failing it, so even an unlucky scale-down mid-stitch self-heals on
the next drain.
83     Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to
the CF proxy
84     (M9) before inviting testers.
85
86     ## 4. M7 acceptance (the owner runlog)
87     Capture a `DEPLOY_REPORT_cloud-serving_<date>.md` under
88     `docs/COMPUTE_DECOUPLED_SERVING/runlogs/` (mirrors the truly-local template): the exact
commands,
89     rally counts, `gpu_active_seconds`/`wall_seconds`/`bytes_out`, a reel smoke-test
(`ffmpeg -f null -`
90     exit 0 + a 5-frame contact sheet), gotchas, and the **$0 teardown audit**.
91
92     ## Gotchas
93     - The dispatched container **must run inline** ? `CloudRunCompute` forces
`compute_target=local_gpu`
94     + `indexing.detector_impl=native` in the per-execution args so the Job never re-
dispatches itself.
95     - Completion is a **bucket marker** (`<out_uri>/_dispatch/<token>.done`), not the Cloud
Run API status
96     ? so it reuses the storage layer + `execution_policy.poll_until` and survives a
control-plane restart.
97     - A **failed** Job (validation rc 2 / segmenter rc 3) now writes a **failure marker**
carrying the
98     real non-zero rc, so the dispatcher's poll terminates **FAST + accurately** (no 30-min
hang on a bad
99     video). Only a **hung/crashed** container (no marker at all) makes the bounded poll
time out
100     (`ExecutionPolicy.max_wait_sec`); the worker CLI maps that to a clean **rc 4**, and
the API path
101     surfaces it as a `cloud dispatch error` job failure (not a traceback).
102     - libx264 encode (parity, D16) + NVDEC decode-only; the image keeps ffmpeg's default
libx264.
103     - **`GOOGLE_CLOUD_PROJECT` is required** for the real client ?
`GoogleCloudRunJobClient.run_job`
104     fails fast with a clear `RuntimeError` if it is unset/empty (it must resolve a real
`job_path`;
105     an unset project would otherwise build a bogus `projects/None/...` name). Export it
(step 3) before
106     dispatching. The guard runs before the `google-cloud-run` import, so it is exercised
in CI even
107     though the SDK isn't a CI/test dependency.
108     - **Hand-rolled `jobs execute --args` with `gs://` URIs:** gcloud's default comma-split
plus its
109     alternate-delimiter syntax both bite ? repeated `--set` tokens trip the comma parser,
and a
110     `^:^` delimiter splits the `:` inside `gs://`. Use a delimiter that appears nowhere in
the args,
111     e.g. `--args='^@@^-v@@infer@@-video-uri=gs://?'`. (The dispatcher path doesn't have
this
112     problem ? prefer it; this is for one-off diagnostics.) Container-arg overrides must
include the
113     dispatcher's two forced `--set`s: `deployment.compute_target=local_gpu`,
114     `indexing.detector_impl=native`.
115     - **Git Bash (MSYS) on Windows mangles Unix-looking values** before gcloud sees them:

```

```

116     `--add-volume-mount ?mount-path=/gcs` became `C:/Program Files/Git/gcs`, and
117     `--command=/bin/bash` became `C:/Program Files/Git/usr/bin/bash` (the container then
exec-fails).
118     Run gcloud commands with such flags from PowerShell (`MSYS_NO_PATHCONV=1` breaks
gcloud's own
119     wrapper). PowerShell has its own trap: quote any flag value containing commas.
120     - **First execution after an image push** spends minutes in "Container image import in
progress"
121     (visible in `executions describe status.conditions`, NOT in logs) ? that's progress,
not a hang;
122     later executions reuse the imported image.
123

```

14. user (2026-07-04T02:26:18.833Z)

```

1      # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2
3      > **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** .
**Owner:** Avi . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; pointer in memory
`current-state` + `docs/NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) $3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
7      > **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed
**compute-decoupled serving** because **truly-local is a co-equal, first-class profile**, not an
afterthought.
8      > **Honesty note:** claims tagged `[verified]` (checked against code by the design
workflow `wf_56274ff0`), `[design]` (proposed), `[researched]`.
9
10     ---
11
12     > **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video
? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next
to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user
*confirms* (D15).
13
14     ## 0. TL;DR
15     The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three
deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch
on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process
vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB
/ mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local**
(local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive /
bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus
**cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the
same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the
**parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.
16
17     ## 1. Problem & goal
18     - **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4)
and is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero +
cost guard)** become this project. The owner needs **both** a hosted service *and* a truly-local
mode, with the **core (detect+stitch) unaffected** by which one runs.
19     - **The two user-facing outcomes** (owner, 2026-06-21):
20     - ***(a) Hosted service** ? operate on small **local uploads (`<2GB`)** *and* load from
**gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching**
(stitching is compute-intensive ? run it where its cost is accounted for).
21     - ***(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local
GPU**, run inference *and* stitching **completely locally**.

```


22 - *****"Good" looks like:**** the same `video ? rallies ? reel` core gives identical windows
+ stitched ranges whether it runs on a laptop or Cloud Run; switching is a ****config/startup
choice****, never a code branch in the core.

23

24 **## 2. Decisions locked**

25

	#	Decision	Source	Implication
26		#		Decision Source Implication
27		---		----- ----- -----
28		D1		The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile branch inside it** . design `[verified]` All profile difference lives in config + the two registries.
29		D2		**Two registries:** `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`. ADR-014 Realizes PLATFORM_ARCHITECTURE's `local`/`hosted`/`byo_worker`.
30		D3		**Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred. ADR-014 Enum/seam reserved so BYO isn't precluded.
31		D4		In **cloud-serving, STITCH runs on Cloud Run** (co-located with detect) so CPU/wall + egress are metered into the per-job cost ledger. owner (a) Stitch-on-laptop would hide real cost.
32		D5		**Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory); **detect** is byte-exact only same-GPU. deploy report 2026-06-20 Cross-GPU asserted at the rally/window level, never CSV bytes.
33		D6		**Default-OFF, smallest-safe-first, one PR per milestone** ; any core-caller change (e.g. configurable output base) ships behind a flag + a Launch Report. [[launch-report- convention]] Zero-regression discipline.
34		D7		**Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that ~\$500/mo idle would break per-video billing). owner 2026-06-21 M6/M7. Revisit a warm lane only on a UX complaint.
35		D8		**Pricebook = a versioned DB table** ; cost computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore market); re-versioned when GCP prices change. owner 2026-06-21 M8.
36		D9		**Edge auth = Cloudflare Access** (reuse the site's existing CF). Lock the Cloud Run **ingress to the CF proxy** + **verify the `Cf-Access` JWT signature** (so a direct hit to the Cloud Run URL can't spoof the identity header). owner 2026-06-21 M9. One identity system.
37		D10		**Free-tier "data-for-reels" trade:** free reels in exchange for **training rights** (uploaded footage + derived trajectories/labels); **paid tier = no-train** (privacy is the paid feature). **Carve-outs:** train ONLY on attested **ADULT** footage (minor footage ? reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on **DERIVED** data + **auto-delete raw** ; **ToS counsel-reviewed** ; live training-on-uploads **gated to M9** (public signup). Truly-local needs no DPA. owner 2026-06-21 M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel.
38		D11		**Quality floor: Gemini handoff ON** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report** ; the owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) owner 2026-06-21 M7 + D12.
39		D12		**Data-flywheel harness is IN SCOPE (owner add, Q5): **capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones to the golden TRAINING set** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). owner 2026-06-21 new **M11** ; ties D10 (consent) + `data_pipeline/` + the eval/experiment harness.
40		D13		**Profile is user-switchable ANYTIME ? a headline FEATURE.** The same user runs **local today, cloud tomorrow, back to local** ? it's just a config/startup choice; the pure core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ? your call, switch anytime"). owner 2026-06-21 Honest caveat: equivalent at the **rally/window** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is **between** jobs.
41		D14		**NORTHSTAR ? operate toward this:** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source** . owner 2026-06-21 Implies a **`gdrive-api` (OAuth) StorageBackend** (read source + write reel back) ? distinct from the local **mount** backend; **resolves \$8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive

write); not necessarily v1, but the design must not preclude it (the `StorageBackend` ABC accommodates it). |

42 | D15 | ****Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.**** A GPU owner may still choose cloud; ****cloud (= spend) is never entered silently****; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends §4.3 (auto-detect = a proposed default, not an auto-decision). |

43 | D16 | ****M6/M7 implementation SHAPE + the cheap §8 picks (owner 2026-06-21):**** (a) ****Cloud Run JOBS**** (not Services-with-GPU) for the detect+stitch unit, ****bucket-poll** for completion** (reuses `JobWaiting`/`claim\_due\_job` verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) ****stitch ENCODE = libx264**** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? ****NVDEC for DECODE only**** (a speed lever that never touches output bytes); ****NVENC encode is DEFERRED behind a config knob****, revisit only if M7 cost data justifies relaxing parity (§8 Q8); (c) ****input cap = hard-reject `<2GB`**** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? §8 Q9). | owner 2026-06-21 | M6/M7. Resolves §8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |

44 | D17 | ****Real cloud verification BUDGET = \$100 / ?10k cap (owner 2026-06-21):**** the owner authorized actual GCP spend for due-diligence + production-grade testing/verification of M5?M7 (real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), ****provided the cap is respected**** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and ****DELETE every resource after**** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |

45

46 **## 3. Foundation ? evolution / ADR lineage ? *trace the idea end-to-end***

47 This project is the latest step in a long decision chain; each ADR contributed a piece and a ****subtlety that carries forward****:

48

49 | ADR / doc | Contributed | Subtlety carried into this project |

50 |---|---|---|

51 | ****ADR-005**** | Permissive licensing (MIT/Apache-2.0/BSD only) | The engine + any served model + the ****container image**** (ffmpeg, fonts, weights) stay commercial-clean. |

52 | ****ADR-008**** | Owned-model topology; ****train==serve** featurizer single-sourcing** | The "core unaffected across profiles" is the same parity discipline, extended from train/serve to ****local/cloud****. |

53 | ****ADR-009**** | KhelSutra Guru; ****local-first delivery**** | The ****truly-local profile**** is the direct continuation ? privacy/offline self-host stays first-class. |

54 | ****ADR-010**** | DECOUPLED: D0?GCP pivot | The cloud-compute origin (GPU + co-located bucket). |

55 | ****ADR-011**** | DECOUPLED scope = M0?M2+M3.5; ****deferred M3 (serving) + M4 (billing/cost-guard)****; override the ****rent nothing / not a service**** posture | ****This project realizes M3/M4.**** Carries §06's owner-gated gates: ****DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3****. |

56 | ****ADR-012**** | GPU-native; TPU deferred; ****Cloud Run+GPU scale-to-zero = serving model****; ****NVDEC**** named as the decode lever | The serving substrate + the 7-step seed scope this expands. |

57 | ****ADR-013**** | Drop D0; ****GCP sole compute stack****; \$200 D0 credits ? non-compute only | The provider for cloud-serving; D0 Spaces remains only a possible S3 ****storage* target**. |

58 | ****? ADR-014 (new)**** | ****Compute-decoupled serving:**** one pure core, profile-selected compute+storage; stitch-where-metered | This doc. ****Refines**** ADR-011/012, does ****not**** supersede them. |

59 | PLATFORM_ARCHITECTURE.md | The `ComputeBackend`/`StorageBackend` registries | The abstraction this realizes in code. |

60 | VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md | L0 fully-local tier / `LocalDesktop` backend | The truly-local profile = these, productionized. |

61 | 07-COMPUTE-ABSTRACTION (archived) | `DeviceContext`/`DeviceProfile` (designed) | WASB hardcodes `device='cuda'` with ****no CPU fallback**** ? a portability gap M4 closes. |

62 | DEPLOY_REPORT_GCP_2026-06-20 (archived) | First real L4 run; ****cross-GPU sub-pixel parity finding**** | Why D5 (parity at window level, not CSV bytes). |

63

64 **## 4. Design / architecture**

65 See [`architecture.svg`](architecture.svg) for the picture. ****The core never changes; the box around it does.****

66

67 **### 4.1 The core contract (must stay byte/behaviour-identical) `[verified]`**

```

68     1. **DETECT (GPU):** `DetectorRunner.run_predict(video_path, output_dir, video_id) ?
CSV[Frame,Visibility,X,Y]` (`backend/pipeline/detectors/base.py:164`). Three interchangeable
impls ? `WasbRunner` (WSL), `NativeWasbRunner` (Linux GPU), `StubDetectorRunner` (CPU mock) ?
emit the **identical CSV schema**. The trajectory CSV is the stable artifact boundary.
69     2. **WINDOW (CPU pure fn):** `trajectory_to_action_windows(points, fps, frame_width, ?)
? windows` (`tracknet_runner.py:281`). A shared @staticmethod, zero GPU/IO/randomness ? same
trajectory + config = byte-identical candidate windows. Used by every runner *and* the
eval/regression stack verbatim.
70     3. **STITCH (CPU/ffmpeg):** `VideoCompiler.compile_highlights(raw_video, segments,
out_name) ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffmpeg ? watermark
? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.
71
72     **Non-goal (the trap to avoid):** *no* code branch inside detect/window/stitch keyed on
profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-
identical to CONTROL).
73
74     ### 4.2 The deployment profiles
75     | Profile | Compute | Storage | Orchestration | When |
76     |---|---|---|---|---|
77     | **truly-local** | `compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto`
(WSL); stitch **in-process** | `local` (or `gdrive` = mounted Drive) | `python -m backend.main`
or `worker infer` CLI; existing async job worker | Self-host / offline / privacy / dev-on-a-GPU-
box ? owner (b) |
78     | **cloud-serving** | `compute_target=cloud_run`; `detector_impl=native` (L4, cuda);
**detect AND stitch on the same Cloud Run job** | `gcs` (or gdrive mount / assembled upload);
per-video workspace; reels via signed URL | Cloud Run control plane enqueues ? Cloud Run GPU job
pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero | Hosted
SaaS ? owner (a) |
79     | **cpu-mock** | `compute_target=cpu_mock`; `detector_impl=stub` (synth/replay) |
`local`/in-memory fakes | `run_all_tests.py` / pytest | CI, regression, profile-parity, GPU-free
dev |
80     | *byo_worker* (deferred) | tenant GPU via outbound pull agent | tenant's own bucket
(signed URLs) | long-poll pull; same job table scoped by tenant | Privacy-sensitive enterprise ?
**DEFERRED**, enum/seam reserved |
81
82     ### 4.3 Profile selection (the "startup/setup config") `[design]`
83     ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-
detect in `load_config`:
84     - `deployment.profile ? {truly-local | cloud-serving | cpu-mock}` (empty = today's
manual knobs, **fully backward-compatible**).
85     - Selecting a profile applies a **coherent preset bundle** of *existing* knobs that
today drift independently: **INPUT** (new `input_backend`/`input_root`, parallel to the output-
side `StorageConfig`), **COMPUTE** (new `compute_target` enum locking `detector_impl` +
`skip_ai_handoff` + workspace strategy), **STORAGE**
(`storage.backend`/`workspace_root`/`single_folder_per_video`, all already exist).
86     - **Precedence:** built-in defaults ? profile preset ? `config.json` ? env
(`RALLY_DETECTOR_IMPL`, `WASB_*`, `DEPLOYMENT_PROFILE`, ?) ? worker `--set k=v`. Every knob
stays individually overridable.
87     - **Env auto-detect SUGGESTS, the user CONFIRMS (D15 ? no surprise execution): ** the env
gives a *proposed* default (`K_SERVICE` ? cloud-serving; `CI=true` ? cpu-mock; else
`torch.cuda.is_available()` ? truly-local), but the user **explicitly confirms or overrides** it
? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The
active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast**
(cloud-serving without GCS creds/GPU ? loud error before any job).
88     - The worker (`cmd_infer`/`cmd_train`) is **already profile-agnostic** (reads
`config.storage`, accepts `--config`/`--set`) ? no worker rewrite.
89
90     ### 4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
91     DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In
**cloud-serving, stitch is co-located in the Cloud Run job ** so `gpu_active_seconds` (detect) +
`wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ?
the per-job cost ledger. Running stitch on a laptop would leave that cost
**invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in
the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get
bursty ? the job table + `claim_due_job` already supports both.)

```

92

93 ### 4.5 Reuse map ? most of the seams already exist `[verified]`

94 **Exists:** the 3-stage core; `\_resolve\_runner` (the single compute seam);
`StubDetectorRunner` (`replay\_csv` = byte-exact \$0 anchor); `StorageBackend` ABC + 4 backends +
`StorageRef.parse\_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control
plane (`jobs` table, `claim\_due\_job` CAS); `ExecutionPolicy WAIT\_AND\_RESUME`/`JobWaiting`
(carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip\_ai\_handoff`
(LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial:** per-video
workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user\_id` columns
exist). **Net-new:** the `deployment` section; `input\_backend`; configurable output base (kill
hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-
auth header extraction; startup env detection.

95

96 ### 4.6 The data-flywheel harness (D12 / M11) `[design]`

97 Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip
Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline
additionally:

98 1. **Captures** the source upload + the Gemini reel + the rally output
(intervals/report) into the per-video workspace (`workspaces/<video\_id>/`), instead of the no-
consent auto-delete path.

99 2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw
video kept only for the training pool, not the no-consent path).

100 3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini
(`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks
the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").

101 4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING
rows** (human-reviewed labels via the annotation flow ? `data\_pipeline/`), growing the corpus ?
the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.

102 This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the
PERSONALIZATION contribution model): every consented free reel makes the owned model better.
Reuses `data\_pipeline/`, `backend/eval/`, and the M9 consent gate; **adults-only**, raw-
retention gated by the D10 consent.

103

104 ## 6. Honest limits ? what this does NOT do / prove

105 - A green CI suite proves **plumbing + determinism (window+stitch) + profile-parity**
for \$0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-
side, real-video).

106 - **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a
CSV-byte claim.

107 - The cloud `0.1.0-l4` weights are **n=2 plumbing-grade** ? cloud-serving ships with
Gemini handoff ON until the owned model clears a floor (open Q).

108 - This does not resolve **WASB-C3** (sharing a trained model) or the **DPA/consent**
gate ? both remain owner/counsel-gated before public, non-owner serving.

109

110 ## 7. Deliverables & progress tracker ? **source of truth**

111 Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off
`origin/master`, no stacking. Default-OFF where it touches core callers.

112

	ID	Deliverable	Depends	Gated?	Status	PR
	M0	This design doc + project dir + `runlogs/` + ADR-014			?	owner sign-off
? APPROVED 2026-06-21						
	M1	`deployment.profile` + `compute_target` config + preset bundles + precedence/auto-detect; unit tests (no behaviour change, profile=' default)			M0	safe
		[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)				
	M2	`input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri` (local=identity); storage-fake tests			M1	safe
		[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280)				
	M3	Configurable output/scratch base ? kill hardcoded `output/` (`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**;				golden
		+ stub-E2E byte-identical			M1	?
		[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282)				
	M4	`DeviceContext` + WASB **CPU-fallback**;				unified healthcheck/device wiring
		[#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283)				

120 | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed
runlog (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |

121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
containerized worker image (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |

122 | M7 | **cloud-serving profile** e2e on L4: upload<2GB + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |

123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation +
`/api/.../cost` | M7 | ? owner (billing) | ? | [288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |

124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user\_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |
[290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT
verify + owner_id tag, default-OFF ?; **exposure-safety boot posture** in `startup.py` ? edge-
auth-on requires the `allowed\_video\_root` jail + CF team/AUD or the server refuses to start, see
changelog; consent cols + ToS/DPA ? owner/counsel) |

125 | M10 | `byo\_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit
owner approval | ? | ? |

126 | **M11** | **Data-flywheel harness (D12)**: on a consented adult job, **capture** the
upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-
eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data\_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | **skeleton landed** (see changelog):
`backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `\_maybe\_run\_flywheel` hook,
consent faked-deny; live activation ? owner/counsel (consent schema) + owned-model/golden
data |

127 | **M12** | **gdrive-api** (OAuth) StorageBackend (D14 Northstar): cloud reads the
source from + writes the stitched reel **back** to the user's Google Drive next to the source
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? |
backend landed (see changelog): `backend/storage/gdrive\_api.py` `GDriveApiStorage` +
`gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app** ?
owner** |

128

129 **Testing strategy is a first-class deliverable** ?
TESTING_STRATEGY.md. **Run logs / sample runs for posterity** ?
runlogs/.

130

131 **## 8. Open questions ? owner** (blocking-gates vs nice-to-knows)*

132 **?? The 5 gating questions are LOCKED** (owner, 2026-06-21) ? see §2 D7?D12:**

133 1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).

134 3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).

135 4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to
proxy).

136 5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data;
counsel-gated to M9).

137 6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until
the owned model clears the floor; capture?store?eval?goldenize meanwhile).

138

139 **?? The remaining smaller picks are LOCKED** (owner, 2026-06-21) ? see §2 D16:**

140 2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the
job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).

141 7. ? **RESOLVED ? D14**: cloud needs a **gdrive-api** (OAuth) StorageBackend (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.

142 8. ? **Codec ? D16**: **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only**
(speed, no output-byte change); NVENC-encode deferred behind a config knob.

143 9. ? **Input cap ? D16**: **hard-reject <2GB** for v1, as a config cap (raise to 4GB

later in one line).

```

144     10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the
name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.
145
146     ## 9. Definition of done
147     - [ ] M1?M4 land (config + input + output-base + device layer), each green on golden
cross-OS + experiment no-regression.
148     - [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical
windows + segment ranges.
149     - [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog /
DEPLOY_REPORT** with a real sample run.
150     - [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth
(M9) gates multi-tenant.
151     - [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented
adult jobs; the owned model is on a measured path to the D11 floor.
152     - [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.
153
154     ## 10. References
155     **ADR lineage:** ADR-005/008/009/010/011/012/013 ? **ADR-014**
(`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`,
`HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`,
`LOCAL_APPLIANCE_ARCHITECTURE.md` (khelsutra). **Archived predecessor:**
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-
ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code
anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`,
`stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`,
`backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`,
`backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-
mappers ? synthesis; 58 findings).
156
157     ---
158     ### Changelog
159     - 2026-06-21 ? created from the design workflow output; M0 in progress.
160     - 2026-06-21 ? **M0 APPROVED**; **M1 landed**
([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)):
`deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion
(D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core
change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach).
Suite 1068 green, cov 84.11%.
161     - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/280)): `input_backend`/`input_root` + URI-aware main API/CLI input via
`resolve_input_uri`/`acquire_local_input` (local = identity passthrough, byte-for-byte
unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted
on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report
provenance. Suite 1105 green, cov 84.84%.
162     - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/282)): configurable output/scratch base via
`backend/utils/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under
`output.output_dir` (default `"output"`, byte-identical; default-OFF ? no Launch Report). Fixes
the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
163     - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"`
(cuda-if-available-else-cpu); explicit `"cuda"` stays strict (no silent slow-CPU downgrade),
default unchanged. `"auto"` resolves before argv (never reaches `wasb_infer.py`); CUDA probe
cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126
green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
164     - 2026-06-21 ? **M5+ batch authorized + design picks locked** (owner): the remaining $8
picks resolved ? **D16** (M6 = Cloud Run **Jobs** + bucket-poll; **libx264 encode + NVDEC
decode-only**); **hard-reject <2GB** input cap; dir name confirmed) and **D17** (real cloud
verification budget = **$100/?10k cap**, delete-after). Owner gave a blanket go to write+merge
the **default-OFF code** for every remaining gated milestone (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each, reserving the gate for the real
run/spend/calibration/counsel. Grounded by gate-map workflow `wf_4849400f`.
165     - 2026-06-21 ? **REAL M7 cloud GPU run ?** (owner-authorized, within the D17 cap):
provisioned an **L4** (`g2-standard-4`, asia-southeast1-a; Mumbai stocked-out) ? native WASB on

```

the L4 ? ****22 rallies**** on the smoke clip ?
`gs://?/outputs/05e0e577?/{intervals.csv,report.json}`; VM DELETED (\$0, ~\$0.21). ? the no-Gemini cloud run needs `--set indexing.skip\_ai\_handoff=true` (the fully-local path emits WASB candidate windows directly ? the long-flagged "0 rallies" cold-start fix). The reel ****stitch**** validated locally (`VideoCompiler` ? 22-segment highlights, 85 MB). DEPLOY_REPORT runlog = the cloud-GPU dry-run guide (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`).

166 - 2026-06-21 ? ****M9-auth code landed**** ([#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)): `backend/api/auth.py` ? Cloudflare Access ****Cf-Access JWT verify**** (RS256: sig+exp+iss+aud against the team JWKS; lazy `PyJWT[crypto]`; injectable JWKS ? fully-offline tests) + `resolve\_tenant` ? tags `create\_job(owner\_id=?)`; `SecurityConfig.{edge\_auth\_enabled=False,cf\_team\_domain,cf\_access\_aud}` + CORS via `RALLY\_CORS\_ALLOW\_ORIGINS`. ****Default-OFF.**** Security review (`wf\_9b4c71d0`) found NO bypass; locked alg=none + HS256-confusion rejection. ****? M9 ? ? the consent cols + ToS/DPA are owner/counsel; the consent migration is now v7 (M8 took v6).****

167 - 2026-06-21 ? ****M8 code landed**** ([#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288)): per-job ****cost ledger**** ? `backend/billing.py` (pure `usage` × pricebook` math, USD internal / INR display) + a **\*\*v6 migration\*\*** (NULLABLE cost columns on `jobs` + a versioned `pricebook` table seeded `placeholder-v0` at \$0; additive-NULL, idempotent) + `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET /api/jobs/{id}/cost` + `api/videos/{id}/cost` + `BillingConfig` (**\*\*default-OFF\*\***) + a gated `_maybe_record_job_cost` hook. Adversarial review (`wf_498b24af`) folded a **\*\*major silent-GPU-under-bill\*\*** (an unpriced `gpu_type` is now surfaced, not invisible \$0) + the ****honest no-op**** caveat (usage emission is a follow-up ? cost 0 until wired) + breakdown reconciliation. Suite green, ruff+mypy clean. ****? M8 ? ? owner inserts real GCP prices (a new pricebook version) + the worker usage-emission is a follow-up; placeholder ships \$0 = no user-facing change. v6 is the shared bump (M9-consent cols extend it).****

168 - 2026-06-21 ? ****M6 code landed**** ([#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run ****Job**** (injectable client) then ****bucket-polls**** a done-marker via `execution\_policy.poll\_until` + the storage layer (D16); `get\_compute\_backend` returns a backend ONLY for `compute\_target=cloud\_run` (else `None` ? the worker's in-process path, ****default-OFF****). The dispatched container runs ****inline**** (forced `local\_gpu`+native, never re-dispatch); the worker writes a `--done-uri` completion marker on success. `deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env; weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf\_87b22fa4`) folded a ****blocker**** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. ****? M6 ? ? the real build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).****

169 - 2026-06-21 ? ****M5 code landed**** ([#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286)): `backend/config/startup.py` ? per-profile fail/warn-fast startup checks (cloud-serving + non-cloud storage = the one fatal coherence error; GPU/creds mismatches = warnings, the runner healthcheck stays the authority), wired into the server (`\_enforce\_startup\_or\_exit` ? exit 1) + the worker (`cmd\_infer`/`cmd\_train` ? rc 2), default-OFF (probe gated behind an active profile ? the worker stays torch-free at `profile=""`). ****L4 profile-parity test**** (`tests/test\_profile\_parity.py`): same stub trajectory ? byte-identical windows + segment ranges across truly-local / cpu-mock / no-profile + worker-vs-in-process. Adversarial 4-lens review folded (`wf\_34671ac1`). Suite green, ruff+mypy clean. ****? M5 ? ? the real-GPU truly-local runlog (L5) is owner-side**** (template pre-staged in `runlogs/`).

170 - 2026-06-21 ? ****M9 hardening: exposure-safety boot posture**** (alpha-shareable "tunnel-to-a-box" enabler): extended `backend/config/startup.py` with a ****server-only exposure posture**** (new `include\_exposure=True` from `\_enforce\_startup\_or\_exit`; the worker stays unaffected ? it reads `--video-uri`, not untrusted tenant paths). Gated on `security.edge\_auth\_enabled` (****default-OFF ? strict no-op****, byte-identical to today): when the owner flips edge auth ON to expose the engine behind the CF-Access gate, three checks fire at ****boot**** ? `EXPOSED\_NO\_VIDEO\_ROOT` (****fatal****: `allowed\_video\_root` jail unset ? a tenant could read an arbitrary local file), `EXPOSED\_EDGE\_AUTH\_INCOMPLETE` (****fatal****: CF team-domain/AUD missing ? lifts the current first-request 401 to boot), `EXPOSED\_AUTH\_EXTRA\_MISSING` (****warn****: `PyJWT[crypto]` not importable ? would 500 per request). So a half-configured exposure fails fast instead of leaking/500-ing. +15 unit tests (default-OFF no-op, worker-ignores, both-fatals-compose-with-profile-checks, server-exit, WARN-only-no-exit). Adversarial 3-lens review (`wf\_35b735a9`) confirmed default-OFF byte-identical + no bypass; folded a ****whitespace-drift**** (aligned `auth.resolve\_tenant` to `.strip()` team/AUD so "configured" means the same at boot + at request time) + named the config fields in the edge-auth-incomplete message. Pairs with the appliance ****T3**** network posture (raw `:8000` refused ? owner-run). Suite green, ruff+mypy clean. ****? backstops M9 ? the actual exposure flip + CF team/AUD/ingress + key rotation stay**

owner-side. (Deferred to the PR-2 runbook: document that `upload\_dir` must sit under `allowed\_video\_root`, and that you MUST set `edge\_auth\_enabled=True` when exposing ? these checks only fire then.)**

171 - 2026-06-22 ? **M11 skeleton landed** (data-flywheel harness, D12):
`backend/flywheel.py` (`capture\_job` ? per-video `VideoWorkspace` via `backend.storage.put`; `shadow\_eval` ? `experiment.compare`/`compare\_unlabeled`; `propose\_promotion` ? `promotion.promote\_if\_served\_safe` with Gemini=control/owned=proposal; `run\_for\_job` orchestrator) + `FlywheelConfig` (**default-OFF**) + a gated `\_maybe\_run\_flywheel` job-completion hook mirroring M8's cost hook. **TWO independent guards keep it INERT in production:** `flywheel.enabled=False` (default) AND `consent\_attested` is a **FAKED hard-deny** (returns False for every job) ? so even enabled it captures **nothing** until the real consent schema lands. This is the PLUMBING only; it reuses the real workspace/eval/promotion/storage code so it works the day consent + the owned model + golden labels are wired. **NO DB migration** (the skeleton persists no consent). +13 tests (default-OFF no-op, enabled-but-consent-denied no-op, consented-capture writes the manifest, delegation of shadow-eval/promote, the hook is best-effort). Suite green, ruff+mypy clean. **? M11 ? ? LIVE activation is owner/counsel (the consent ToS/DPA + schema ? the consent columns) + owner/GPU (the owned model + golden corpus for shadow-eval/promotion). ? the future M9-consent migration is now `v9` (v7 = `locale`, v8 = the compute-picker `compute` column ? the schema ladder drifts as siblings land; check `PRAGMA user\_version` before authoring).**

172 - 2026-06-22 ? **M12 backend landed** (`gdrive-api://` OAuth/API StorageBackend, D14 Northstar): `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` ? the headless **Drive-API** round-trip (read source + write reel back), distinct from the mount-only `gdrive://` (left untouched). Path-keyed `gdrive-api://<path>` resolves each component ? a Drive file id via `files().list`; implements `fetch\_to\_local` (`get\_media().execute()` bytes), `put` (folder-creating `files().create` + a `\_media\_body` seam), `exists`/`stat`/`list`/`delete`. Wired the scheme into `ref.py` (`\_KNOWN` + the path-keyed branch), `\_\_init\_\_.backend\_class`, and the `StorageConfig.backend`/`input\_backend` Literals + a `gdrive\_api` settings dict (read via a **hyphen?underscore** lookup in `get\_storage`). **All Google libs lazy-imported** (`import backend.storage` never pulls them); a `service` is **injectable** ? fully-offline tests via an in-memory fake Drive (round-trip + scheme + registry + the underscore-settings lookup); `\_build\_service` (ADC over the Drive scope) raises a clear `storage-gdrive-api`-extra error when libs/creds are absent. **NO DB migration.** Adversarial 2-lens review (`wf\_ae8c9b0f`). Suite green (1466), ruff+mypy clean. **? M12 ? ? the LIVE per-user interactive OAuth app (consumer "connect your Drive": client id/secret + refresh tokens) is owner-gated; the backend ships the service-account/ADC build + the fake. Reads no secrets from config (creds via ADC only).**

173

15. assistant (2026-07-04T02:26:20.926Z)

Now let me search for any M9-specific documentation and check the pyproject.toml for the auth extra:

16. user (2026-07-04T02:26:22.258Z)

```

1      # PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
2      #
3      # Build backend: hatchling. `backend/` is an intentional NAMESPACE package
4      # (no `backend/__init__.py`; mypy.ini sets explicit_package_bases for the same
5      # reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6      # auto-detection would otherwise miss the un-__init__'d `backend` root.
7      #
8      # torch/torchvision are deliberately NOT listed in [project.dependencies]:
9      # their CPU/CUDA wheels need an out-of-band index (--index-url / --extra-index-url),
10     # which PEP 621 cannot express. Install them separately ? see [project.optional-
11     dependencies]
12     # `gpu` (documentation only) and the README. CI installs CPU torch on its own line.
13     [build-system]
14     requires = ["hatchling"]
15     build-backend = "hatchling.build"
16
17     [project]
18     name = "badminton-highlight-indexer"
19     version = "0.1.0"
20     description = "AI-assisted rally segmentation and highlight indexing for racket sports."

```



```

20     readme = "README.md"
21     requires-python = ">=3.10"
22     license = { text = "MIT" }
23     authors = [{ name = "Avi Dullu", email = "avi.dullu@gmail.com" }]
24     keywords = ["badminton", "sports", "video", "highlights", "rally", "segmentation"]
25
26     # Runtime deps mirror requirements.txt, EXCEPT torch/torchvision (see header).
27     dependencies = [
28         "fastapi>=0.111.0",
29         "uvicorn>=0.30.1",
30         "opencv-python>=4.9.0.80",
31         "numpy>=1.26.4",
32         "pydantic>=2.7.1",
33         "jinja2>=3.1.4",
34         "python-dotenv>=1.0.0",
35         "google-genai>=2.4.0",
36         "supervision>=0.21.0",
37         # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d
/
38         # maximum_filter1d); previously present only TRANSITIVELY via supervision, so a
supervision
39         # bump or a pruned-transitive freeze could silently break inference. Declare it.
(BSD-3.)
40         "scipy>=1.10",
41         # D13/P12: ByteTrack moved to the standalone `trackers` package (Apache-2.0). Its
42         # PyPI wheel is broken (metadata-only), so it MUST be installed from git ? see
43         # requirements.txt. PEP 621 cannot express a git URL, so it's NOT listed here
44         # (same pattern as torch/torchvision). CI installs it on its own line.
45     ]
46
47     [project.optional-dependencies]
48     # Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).
49     dev = [
50         "pytest",
51         "pytest-cov",
52         "httpx",          # fastapi TestClient transport
53         "ruff",
54         "mypy",
55         "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
56     ]
57
58     # M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when
security.edge_auth_enabled=True.
59     # `jwt` is lazy-imported, so the base runtime works without this; install it for
hosted/multi-tenant.
60     auth = ["PyJWT[crypto]>=2.8"]
61
62     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
63     # no-op if absent). Pinned to the canonical private repo's immutable commit
64     # SHA; fetched over HTTPS. If you lack access, simply omit this group ? reporting
65     # tests skip via pytest.importorskip. For local dev against a sibling checkout:
66     # pip install -e ../sports-obsreport --no-build-isolation
67     reporting = [
68         "obsreport @ git+https://github.com/Khelsutra/sports-
obsreport.git@a0c7252c6e4159f8702360bea1cb46e5148b5c9f",
69     ]
70
71     # `gpu` is DOCUMENTATION ONLY ? it is intentionally empty. torch/torchvision
72     # CUDA wheels require `--index-url`/`--extra-index-url`, which PEP 621 cannot
73     # express, so they are NOT declarable here. Install them out-of-band instead:
74     # # CUDA 12.4 box:
75     # pip install torch>=2.12.0 torchvision>=0.27.0 --index-url
https://download.pytorch.org/whl/cu124
76     # # CPU-only box (also what CI uses):
77     # pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu

```

```

78     gpu = []
79
80     # Storage backends (docs/DECOUPLED_COMPUTE/). The default `local` backend needs NONE of
81     # these. Cloud SDKs are imported lazily, so `import backend.storage` works without them;
82     # install the extra for the bucket you use. (boto3's S3 client covers AWS + Cloudflare
R2 +
83     # DO Spaces + MinIO via endpoint_url.) `gdrive://` needs NO extra ? it reads a locally
84     # MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
85     storage-s3 = ["boto3>=1.34"]
86     storage-gcs = ["google-cloud-storage>=2.16"]
87
88     [project.scripts]
89     # `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
90     indexer = "backend.main:main"
91
92     [tool.hatch.metadata]
93     # The `reporting` extra pins obsreport to a git tag (a PEP 508 *direct
94     # reference*); hatchling requires opting in to allow those in metadata.
95     allow-direct-references = true
96
97     [tool.hatch.build.targets.wheel]
98     # `backend` is a namespace package (no __init__.py); list both import roots
99     # explicitly so an editable/wheel install exposes `import backend` and
100     # `import training` regardless of cwd.
101     packages = ["backend", "training"]
102

```

17. user (2026-07-04T02:26:22.550Z)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/tests/test_edge_auth.py
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/tests/___pycache___/test_edge_auth.cpython-313-pytest-9.0.3.pyc

18. assistant (2026-07-04T02:26:24.364Z)

Perfect! Now let me check what's in the test file and look at the API endpoint:

19. user (2026-07-04T02:26:25.448Z)

```

1     """Cloudflare Access edge-auth (COMPUTE_DECOUPLED_SERVING M9, D9).
2
3     Fully OFFLINE: an in-test RSA keypair signs Cf-Access JWTs + builds a fake JWKS, so the
verifier +
4     the tenant resolver are exercised with no network and no Cloudflare. Cardinal invariant:
with
5     ``edge_auth_enabled=False`` the resolver is a strict NO-OP (returns None = today's local
single-user)
6     EVEN when a token is present.
7     """
8
9     import json
10    import time
11
12    import pytest
13
14    pytest.importorskip(
15        "jwt"
16    ) # PyJWT[crypto] (in the [dev] extra); skip cleanly where absent
17    import jwt # noqa: E402
18    from cryptography.hazmat.primitives.asymmetric import rsa # noqa: E402
19    from jwt.algorithms import RSAAlgorithm # noqa: E402
20
21    from backend.api import auth as edge_auth # noqa: E402

```

```

22     from backend.config import AppConfig # noqa: E402
23
24     TEAM = "https://team.cloudflareaccess.com"
25     AUD = "test-aud-tag"
26     KID = "test-kid"
27
28
29     @pytest.fixture(scope="module")
30     def keypair():
31         """An RSA private key + a JWKS carrying its public key (kid=test-kid)."""
32         key = rsa.generate_private_key(public_exponent=65537, key_size=2048)
33         jwk = json.loads(RSAAAlgorithm.to_jwk(key.public_key()))
34         jwk.update({"kid": KID, "alg": "RS256", "use": "sig"})
35         return key, {"keys": [jwk]}
36
37
38     def _token(
39         key, *, email="u@example.com", aud=AUD, iss=TEAM, exp_delta=3600, kid=KID, sub=None
40     ):
41         now = int(time.time())
42         claims = {"aud": aud, "iss": iss, "exp": now + exp_delta, "iat": now}
43         if email:
44             claims["email"] = email
45         if sub:
46             claims["sub"] = sub
47         return jwt.encode(claims, key, algorithm="RS256", headers={"kid": kid})
48
49
50     def _cfg(enabled, *, team=TEAM, aud=AUD):
51         return AppConfig.model_validate(
52             {
53                 "security": {
54                     "edge_auth_enabled": enabled,
55                     "cf_team_domain": team,
56                     "cf_access_aud": aud,
57                 }
58             }
59         )
60
61
62     # ----- verifier (accept)
63
64
65     def test_valid_token_returns_email(keypair):
66         key, jwks = keypair
67         assert (
68             edge_auth.verify_cf_access_token(
69                 _token(key), team_domain=TEAM, aud=AUD, jwks=jwks
70             )
71             == "u@example.com"
72         )
73
74
75     def test_falls_back_to_sub_when_no_email(keypair):
76         key, jwks = keypair
77         tok = _token(key, email=None, sub="user-123")
78         assert (
79             edge_auth.verify_cf_access_token(tok, team_domain=TEAM, aud=AUD, jwks=jwks)
80             == "user-123"
81         )
82
83
84     # ----- verifier (reject
85     paths)

```

```

86
87     @pytest.mark.parametrize(
88         "kw",
89         [
90             {"exp_delta": -10}, # expired
91             {"aud": "wrong-aud"}, # wrong audience (not this CF app)
92             {"iss": "https://evil.example"}, # wrong issuer
93             {"kid": "unknown-kid"}, # kid not in the JWKS
94             {"email": None, "sub": None}, # no identity claim
95         ],
96     )
97     def test_invalid_tokens_rejected(keypair, kw):
98         key, jwks = keypair
99         with pytest.raises(edge_auth.EdgeAuthError):
100             edge_auth.verify_cf_access_token(

```

20. user (2026-07-04T02:26:26.673Z)

```

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/api/auth.py:CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/auth.py:def
resolve_tenant(
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/auth.py:    token
= headers.get(CF_ACCESS_HEADER) or headers.get(CF_ACCESS_HEADER.lower())
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/auth.py:
f"missing {CF_ACCESS_HEADER} header (request did not pass Cloudflare Access)"
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/job_worker.py:
# original /api/process pipeline). Both record their terminal state the same way; only process
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/uploads.py:under
security.upload_dir, then submitted to /api/process by the returned path.
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/uploads.py:
"""Finalize: verify part count, move into place, return {path, video_id} for /api/process."""
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/uploads.py:
"next": "POST /api/process with this path",

```

21. assistant (2026-07-04T02:26:28.701Z)

Let me search for the main API endpoint that uses resolve_tenant:

22. user (2026-07-04T02:26:29.651Z)

```

1     """HTTP API surface for the indexer (FastAPI routers)."""
2

```

23. user (2026-07-04T02:26:29.714Z)

(Bash completed with no output)

24. assistant (2026-07-04T02:26:31.255Z)

Let me search more specifically:

25. user (2026-07-04T02:26:32.313Z)

```

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/api/auth.py:113:def
resolve_tenant(
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/config/startup.py:122:    # FIRST request (auth.resolve_tenant); lift it to B00T
so a misconfigured gate never starts
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py:671:
edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py:743:
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py:904:

```

```
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py:1084:
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
```

26. assistant (2026-07-04T02:26:33.895Z)

Now let me check those routes in backend/main.py:

27. user (2026-07-04T02:26:34.625Z)

```
665         the video by content hash exactly as the training pipeline expects. Byte-compatible
with the VLC
666         plugin + browser extension. A later step (owner-gated) promotes labels into the
durable corpus.
667
668         Edge-auth mirrors /api/process (default-OFF ? no-op locally). ``video_id`` is
validated as a bare
669         filename so an untrusted value can't escape the labels dir (arbitrary write)."""
670         try:
671             edge_auth.resolve_tenant(request.headers, request.app.state.config)
672         except edge_auth.EdgeAuthError as e:
673             raise HTTPException(status_code=401, detail=f"edge auth: {e}") from e
674
675         cfg = request.app.state.config
676         labels_dir = os.path.join(output_base(cfg), "labels")
677         try:
678             name = safe_output_filename(f"{req.video_id}.rallies.csv")
679         except ValueError as e:
680             raise HTTPException(status_code=400, detail=f"invalid video_id: {e}") from e
681         os.makedirs(labels_dir, exist_ok=True)
682         path = os.path.join(labels_dir, name)
683         try:
684             path = resolve_under_root(path, labels_dir)
685         except ValueError as e:
686             raise HTTPException(status_code=400, detail=str(e)) from e
687         with open(path, "w", encoding="utf-8", newline="") as f:
688             f.write(req.csv or "")
689         # Count CSV data rows (a line starting with an integer rally_number) for an honest
ack.
690         num_rows = sum(
691             1 for ln in (req.csv or "").splitlines() if ln[:1].isdigit() and "," in ln
692         )
693         logger.info(
694             "annotations saved: video_id=%s rows=%d -> %s", req.video_id, num_rows, path
695         )
696         return {"status": "ok", "num_rows": num_rows, "csv_uri": f"labels/{name}"}
697
698
699     def _storage_settings(cfg) -> Dict[str, Any]:
700         """The storage config as a plain settings dict (NO secrets ? see StorageConfig)."""
701         sc = getattr(cfg, "storage", None)
702         if sc is None:
703             return {}
704         return sc.model_dump() if hasattr(sc, "model_dump") else dict(sc)
705
706
707     def _asset_dest_uri(medium_uri: str, video_id: str, ext: str) -> str:
708         """The durable, MD5-keyed destination inside the chosen medium:
`<medium>/<video_id><ext>`. """
709         return f"{medium_uri.rstrip('/')}/{video_id}{ext}"
710
711
712     def _medium_local_dir(medium_ref, storage_settings: Dict[str, Any]) -> Optional[str]:
713         """The LOCAL directory a ``put`` into this medium would write into (so a free-space
BLOCK can be
```

```

714         enforced, D11), or ``None`` for a cloud bucket (which is cost/quota-warned, never
blocked).
715         ``local`` ? the folder; ``gdrive`` ? the resolved path under the Drive mount.""
716         if medium_ref.backend == "local":
717             return medium_ref.key or "."
718         if medium_ref.backend == "gdrive":
719             from backend.storage.capabilities import _gdrive_mount_root
720
721             root = _gdrive_mount_root(storage_settings)
722             if not root:
723                 return None
724             rel = medium_ref.key.lstrip("/")
725             return os.path.join(root, *rel.split("/")) if rel else root
726         return None # s3 / gcs ? a bucket has no readable local capacity
727
728
729     @app.post("/api/assets", status_code=201)
730     def create_asset(req: AssetRequest, request: Request):
731         """Store a video into the user's chosen medium and register it as a library asset
(P4, D14).
732
733         SELF-HOST posture only (D7): refused with 403 when ``security.allowed_video_root``
is set ? the
734         hosted alpha jails inputs and has no per-user medium credentials, so it stays
upload-to-local.
735         Edge-auth mirrors ``/api/process``. Exactly one source: an ``uploaded_path`` (a
local file from
736         the chunked upload) OR a ``source_uri`` (a video already at rest in a medium). An
uploaded file +
737         a REMOTE medium ? a durable, MD5-keyed copy is ``put`` into the medium while the
local upload
738         stays as the working copy; a ``local`` medium keeps the upload in place (it's
already on this box);
739         a pasted at-rest URI is registered as-is (no copy, D14b). The asset is keyed by the
whole-file
740         MD5 (``compute_video_id``, D3) so it joins the pipeline by content hash, and appears
in
741         ``GET /api/videos``.""
742         try:
743             owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
744         except edge_auth.EdgeAuthError as e:

```

28. user (2026-07-04T02:26:36.149Z)

```

898         worker drains it via `claim_due_job` (EXECUTION_POLICY.md P3 self-scheduling loop).
899         ""
900         # M9 (D9): edge-auth tenant resolution. DEFAULT-OFF ? owner_id None (the local
single-user,
901         # unchanged). When edge_auth_enabled, the Cf-Access JWT is required + verified (a
missing or
902         # spoofed identity header fails here with 401, before any work) and tags the job's
owner_id.
903         try:
904             owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
905         except edge_auth.EdgeAuthError as e:
906             raise HTTPException(status_code=401, detail=f"edge auth: {e}") from e
907
908         allowed_root = getattr(
909             getattr(request.app.state.config, "security", None), "allowed_video_root", None
910         )
911         try:
912             validate_input_path(req.video_path, allowed_root=allowed_root)
913             # M2: resolve the input once (path or storage URI) ? raises ValueError for an
unsupported
914             # scheme, which we map to a clean 400 (an unknown scheme must never 500).

```

```

915         input_ref = storage.resolve_input_ref(
916             req.video_path, getattr(request.app.state.config, "storage", None)
917         )
918     except ValueError as e:
919         raise HTTPException(status_code=400, detail=str(e)) from e
920     # The local-existence fast-fail applies only to a LOCAL input, and stats the
RESOLVED local path
921     # (input_ref.key ? e.g. local://? stripped to the bare path), not the raw URI
string. A
922     # bucket/Drive URI can't be cheaply os.path.exists()'d ? its existence is verified
when the worker
923     # fetches it (a fetch miss fails the job loudly). validate_input_path above still
runs for every
924     # input: its null-byte guard always applies, and a configured allowed_video_root
(hosted mode)
925     # rejects a non-local URI as out-of-root.
926     if input_ref.backend == "local" and not os.path.exists(input_ref.key):
927         raise HTTPException(
928             status_code=404, detail=f"Video file not found at '{req.video_path}'"
929         )
930     # P2 (D11): a bucket/Drive source is fetched to a scratch copy on the worker; fail
fast with 507
931     # if this box can't hold that copy. Best-effort ? an unstatable remote
(network/SDK) or an
932     # unreadable temp fs skips the guard (measuring must not deny a legit job); the
fetch itself
933     # still fails loudly downstream if space genuinely runs out. Local inputs need no
copy ? skipped.
934     if input_ref.backend != "local":
935         need = _remote_input_size_bytes(
936             input_ref, getattr(request.app.state.config, "storage", None)
937         )
938         require_free_bytes(tempfile.gettempdir(), need, stage="fetch")
939     if req.segmenter and not segmenter_registry.get(req.segmenter):
940         available = list(segmenter_registry.list_available().keys())
941         raise HTTPException(
942             status_code=400,
943             detail=f"Segmenter '{req.segmenter}' not found. Available segmenters:
{available}",
944         )
945
946     job_id = uuid.uuid4().hex
947     if not request.app.state.db.create_job(
948         job_id,
949         req.video_path,
950         req.segmenter,
951         req.player_pool,
952         req.max_frames,
953         owner_id=owner_id,
954         locale=req.locale,
955         compute=req.compute,
956         force=req.force,
957     ):
958         raise HTTPException(status_code=500, detail="Failed to persist job record.")
959     _get_executor().submit(
960         _drain_due_jobs, request.app.state.config, request.app.state.db
961     )
962     return JSONResponse(
963         status_code=202,
964         content={"job_id": job_id, "status": "queued", "poll": f"/api/jobs/{job_id}"},
965     )
966
967
968 @app.get("/api/jobs/{job_id}/cost")
969 def get_job_cost(job_id: str, request: Request):

```

```

970         """Per-job cost (COMPUTE_DECOUPLED_SERVING M8, D8). ``cost_usd`` is the source of
truth (matches
971         GCP billing); ``cost_inr`` is a display conversion at the configured FX rate.
``cost_usd`` is
972         null until billing records it (default-OFF / the placeholder pricebook ? 0)."""
973         cost = request.app.state.db.get_job_cost(job_id)
974         if cost is None:
975             raise HTTPException(status_code=404, detail=f"Unknown job_id '{job_id}'")
976         fx = request.app.state.config.billing.fx_inr_per_usd
977         usd = cost.get("cost_usd")

```

29. user (2026-07-04T02:26:36.151Z)

```

738         stays as the working copy; a ``local`` medium keeps the upload in place (it's
already on this box);
739         a pasted at-rest URI is registered as-is (no copy, D14b). The asset is keyed by the
whole-file
740         MD5 (``compute_video_id``, D3) so it joins the pipeline by content hash, and appears
in
741         ``GET /api/videos``.
742         try:
743             owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
744         except edge_auth.EdgeAuthError as e:
745             raise HTTPException(status_code=401, detail=f"edge auth: {e}") from e
746
747         cfg = request.app.state.config
748         sec = getattr(cfg, "security", None)
749         if getattr(sec, "allowed_video_root", None):
750             raise HTTPException(
751                 status_code=403,
752                 detail=(
753                     "storing into a custom medium is disabled on this hosted deployment "
754                     "(security.allowed_video_root is set); upload to local instead."
755                 ),
756             )
757         if bool(req.uploaded_path) == bool(req.source_uri):
758             raise HTTPException(
759                 status_code=400,
760                 detail="provide exactly one of 'uploaded_path' or 'source_uri'.",
761             )
762         if req.uploaded_path and not req.medium_uri:
763             raise HTTPException(
764                 status_code=400, detail="'medium_uri' is required when uploading a file."
765             )
766
767         storage_settings = _storage_settings(cfg)
768
769         # 1) Resolve a LOCAL working copy of the source + its whole-file MD5 (the join key,
D3).
770         if req.uploaded_path:
771             try:
772                 validate_input_path(req.uploaded_path, allowed_root=None)
773                 src_ref = storage.resolve_input_ref(req.uploaded_path, storage_settings)
774             except ValueError as e:
775                 raise HTTPException(status_code=400, detail=str(e)) from e
776             if src_ref.backend != "local" or not os.path.isfile(src_ref.key):
777                 raise HTTPException(
778                     status_code=404,
779                     detail=f"uploaded file not found at '{req.uploaded_path}'",
780                 )
781             local_working = src_ref.key
782         else:
783             try:
784                 validate_input_path(req.source_uri, allowed_root=None)
785                 src_ref = storage.resolve_input_ref(req.source_uri, storage_settings)

```



```

786         except ValueError as e:
787             raise HTTPException(status_code=400, detail=str(e)) from e
788         # A remote at-rest source is fetched to scratch to hash it (D3) ? reserve that
scratch (P2).
789         if src_ref.backend != "local":
790             need = _remote_input_size_bytes(src_ref, getattr(cfg, "storage", None))
791             require_free_bytes(tempfile.gettempdir(), need, stage="fetch")
792         try:
793             local_working = storage.acquire_local_input(
794                 req.source_uri, getattr(cfg, "storage", None)
795             )
796         except ValueError as e:
797             raise HTTPException(status_code=400, detail=str(e)) from e
798         except Exception as e: # noqa: BLE001 ? a fetch miss is a 502, surfaced not
swallowed
799             raise HTTPException(
800                 status_code=502, detail=f"could not resolve source_uri: {e}"
801             ) from e
802         if not os.path.isfile(local_working):
803             raise HTTPException(
804                 status_code=404, detail="source_uri did not resolve to a readable file."
805             )
806
807         cleanup_raw = req.source_uri or ""
808         try:
809             video_id = compute_video_id(local_working)
810             ext = os.path.splitext(local_working)[1] or ".mp4"
811
812             # 2) Decide the durable location + copy only when needed.
813             copied = False
814             if req.source_uri:
815                 stored_uri = (
816                     req.source_uri
817                 ) # already at rest in its medium ? no copy (D14b)

```

30. assistant (2026-07-04T02:26:37.846Z)

Let me check the health endpoint to see if it's exempt:

31. user (2026-07-04T02:26:39.171Z)

```

314:@app.get("/api/health")
1586:         args=(url, url + "api/health"),

```

32. user (2026-07-04T02:26:41.297Z)

```

314     @app.get("/api/health")
315     def health(request: Request):
316         """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
317         (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
318         confirm the engine is reachable and which detector is wired. Never raises."""
319         cfg = request.app.state.config
320         sport = getattr(cfg, "sport", None)
321         indexing = getattr(cfg, "indexing", None)
322         return {
323             "status": "ok",
324             "sport": sport,
325             "default_segmenter": getattr(indexing, "default_segmenter", None),
326         }
327
328
329     # P1: the bundled SPA pins this ? the contract version of the /api surface. BUMP when an
existing
330     # endpoint's request/response shape changes incompatibly (purely additive endpoints need

```

```

no bump).
331     API_VERSION = 1
332
333
334     def _engine_version() -> str:
335         """Installed package version (wheel/editable); 'unknown' if not importable as a
distribution."""
336         try:
337             return _pkg_version("badminton-highlight-indexer")
338         except PackageNotFoundError:
339             return "unknown"
340
341
342     @app.get("/api/version")
343     def api_version():

```

33. assistant (2026-07-04T02:26:43.205Z)

Perfect! Now let me check how the server startup and initialization works with the edge auth:

34. user (2026-07-04T02:26:44.438Z)

```

1206: def _enforce_startup_or_exit(config) -> None:
1209:     the exposed server (`include_exposure=True`, M9 hardening), a half-configured
exposure (edge
1215:         enforce_startup_checks(config, logger=logger, include_exposure=True)
1396:     _enforce_startup_or_exit(

```

35. user (2026-07-04T02:26:46.854Z)

```

1206     def _enforce_startup_or_exit(config) -> None:
1207         """M5: run the startup checks for the server path and `sys.exit(1)` on a FATAL
finding BEFORE
1208         any work ? a fatal incoherent profile (e.g. cloud-serving pointed at local-only
storage) OR, for
1209         the exposed server (`include_exposure=True`, M9 hardening), a half-configured
exposure (edge
1210         auth on without the `allowed_video_root` jail / CF team+AUD). A strict no-op when
no
1211         `deployment.profile` is selected AND edge auth is off (today's behaviour). No GPU
probe here ?
1212         the server stays torch-light; the detector healthcheck is the GPU authority.
Extracted so the
1213         fail-fast wiring + the default-OFF no-op are unit-testable."""
1214         try:
1215             enforce_startup_checks(config, logger=logger, include_exposure=True)
1216         except StartupCheckError as e:
1217             logger.error("[STARTUP] refusing to start: %s", e)
1218             sys.exit(1)
1219
1220
1221     def _maybe_sweep_gemini_orphans(config) -> None:
1222         """01/P11: best-effort sweep of Gemini File-API uploads orphaned by a previous
crashed run.
1223         A no-op unless the configured provider is Gemini AND its API key is present. Never
raises ?
1224         a failed sweep is logged and swallowed. The Gemini client is built with an HTTP
timeout so
1225         a hung connection can't stall startup indefinitely. Extracted for unit tests."""
1226         try:
1227             provider_name = getattr(config, "ai_provider", "")
1228             if provider_name != "gemini":
1229                 return
1230             api_key = os.environ.get("GEMINI_API_KEY")

```

```

1231         if not api_key:
1232             return
1233         provider = provider_registry.get(
1234             "gemini", api_key=api_key, model_name=config.ai_model
1235         )

```

36. assistant (2026-07-04T02:26:48.757Z)

Perfect! Now let me check the deployment docs for CF/DNS setup:

37. user (2026-07-04T02:26:50.123Z)

<persisted-output>

Output too large (33.8KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\90c0b6b8-7430-4870-98cb-e487eacfa1d7\tool-results\behnfj1vc.txt

Preview (first 2KB):

```

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ANALYZERS_AND_RECONCILERS.md: the "signal-not-verdict" rule are domain-
independent. Any temporal problem with **weak, multi-modal cues
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ANALYZERS_AND_RECONCILERS.md:**In-domain prior art that already hard-codes our
cues** (so our delta is the *no-branch discipline*, not the
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ANALYZERS_AND_RECONCILERS.md:- ? **"Signal not verdict" as a deliberate *design-
philosophy delta*** vs. the in-domain prior art that
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ANALYZERS_AND_RECONCILERS.md:- **Detector domain gap inherited downstream** ?
WASB/TrackNet/MonoTrack are trained on other rigs;
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/decisions/DECISIONS.md:1. **Product name: KhelSutra Guru**; domain
**khelsutra.guru** (GoDaddy receipt: 3-yr
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/decisions/DECISIONS.md: registration + Full Domain Protection, renews
2029-06). Earlier "khelstura" spellings were typos.
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/decisions/DECISIONS.md:4. Alpha topology per `docs/HOSTING_PLAN.md`: free
edge (Cloudflare Pages/Access/Tunnel) + the
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/archives/decisions/DECISIONS.md:- Alpha auth = Cloudflare Access JWT at the edge;
the backend reads the authenticated email;
...
</persisted-output>

```

38. assistant (2026-07-04T02:26:52.276Z)

Let me look at the HOSTING_PLAN more carefully:

39. user (2026-07-04T02:26:53.331Z)

```

14:**Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU box)
at
15:**$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker** (Cloud
Run +
19:Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the alpha
needs
30:Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
32: ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on the
box)
41:- **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
42: `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-user
scoping.
47:- **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost (~$0.05/match) and

```

```

54:                                     ??? api.khelsutra.guru ??? Cloud Run (FastAPI
control plane, scale-to-zero)
64:  bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs), Cloud
Run
67:  later); Access allowlist retires.
68:-  Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-to-
zero) .
80:|  **All-Cloudflare** (Pages+Workers+R2) | ~$0 ? ~$5 | Zero egress on R2; cheapest forever |
FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP credits |
95:|  Access | Scope | Used for |
97:|  **Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS>Edit,
Account:Pages>Edit, Account:Access:Apps>Edit, Account:Tunnel>Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
100:|  **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin, Storage
admin, Cloud SQL admin) | Control-plane deploys |
103:API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never live
in the
108:Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate

```

40. assistant (2026-07-04T02:26:55.474Z)

Excellent! Now let me check if there's any existing M9 documentation for the Access app configuration:

41. user (2026-07-04T02:26:56.911Z)

```
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:The audit began by reading the mandated entry points (`docs/README.md`, `docs/CODE_MAP.md`, `CLAUDE.md`) followed by core modules (main.py, config, database, registries, segmenters, detectors, validators, stitcher, providers, reporting, results, etc.) and cross-cutting searches (hardcoded paths, config access patterns, Result usage, prints, registration side-effects).
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:Engine repo (Python, ~1500 tests, CI trusted green on master at e410136). Security/config posture is largely healthy: deploy/gcp/buckets.env single-source (#410) is properly wired into every deploy script, no literal secrets anywhere in the tree (grep for AIza/sk-ghp/_private keys = zero), redaction and edge-auth (CF Access JWT, default-off) are carefully built, startup exposure checks (M9) cover jail/team+AUD/auth-extra/host-allowlist, CI has copyleft license-fail lanes and no pull_request_target/secrets usage, and .gitignore is UTF-8 with artifacts/, output/, *.db all ignored and untracked. Remaining debt clusters in four places: a wildcard CORS default that undermines the localhost-appliance DNS-rebinding guard, the still-open #327 path-jail-vs-gs:// ingest gap, supply-chain pinning (mutable git tag duplicated in 5 files, no lockfile), and small hygiene items (tracked scratch/ files, retired-DO deploy dir still live-coupled).
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- evidence: NEXT_STEPS.md:12-13 ('Where we are (updated 2026-07-01)') still says the site is 'live + browsable on the DO droplet -> http://168.144.126.176:8787'; deploy/DEPLOY_ALPHA.md:3 says 'Status: LIVE (deploy live-verified on the droplet 2026-06-22)' and :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE (2026-06-22)'. Project memory (MEMORY.md:63, session-handoff.md:19): droplet claudе-do-rally-test/168.144.126.176 RETIRED+deleted 2026-06-25; alpha paused pending Cloud Run. nslookup shows api.app.khelsutra.guru still resolve via Cloudflare to a tunnel with no origin (pending CNAME/Access cleanup).
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- evidence: deploy/DEPLOY_ALPHA.md:3 '**Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22...)' ;:35 names 'the CPU droplet 168.144.126.176' as the engine box. The droplet was retired 2026-06-25 (khelsutra commit 5ffb750 #117: 'retire the live-parity monitor ? droplet mirror decommissioned'); grep -i 'retired|decommission' over the repo's docs = 0 hits, so no doc records it.
deploy/GPU_VM_SERVING.md $0 tells the operator to copy the CF-Access AUD from 'the droplet's
```

/srv/khelsutra/config.json' and the tunnel token from 'systemctl cat cloudflared' on the droplet, and calls the droplet a 'hot fallback' ? all impossible now. Secondary rot: DEPLOY_ALPHA.md:143 claims the SPA 'shows a friendly engine offline banner (/api/health ping)' but web/src contains zero references to /api/health (grep -i health = no matches); and open issue #114 (Access sign-in broken) + #97 (split-origin breaks CF Access) contradict the runbook's shipped topology.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- proposed fix: One doc PR: flip DEPLOY_ALPHA.md status to RETIRED/HISTORICAL with a pointer to the retirement date and to issues #97/#114; rewrite GPU_VM_SERVING.md prerequisites to source the Gemini key/AUD/tunnel token from their canonical stores (Drive / CF dashboard) instead of the dead droplet; delete or soften the unimplemented /api/health-banner claim. The related api.khelsutra.guru CNAME + CF-Access application cleanup is Cloudflare-dashboard work (owner-gated) ? note it in the doc as pending.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- verifier notes: Two refinements. (a) "all impossible now" is slightly overstated: per 5ffb750's message the droplet's data was archived to gs://khelsutra-rally-corporus/archive/droplet-claude-do-rally-test-20260625/, so /srv/khelsutra/config.json (and its cf_access_aud) is likely recoverable from that archive, and the tunnel token is recoverable from the Cloudflare dashboard since the tunnel is token-managed (GPU_VM_SERVING.md:24-26). The doc rewrite should point at the gs:// archive + CF dashboard as the canonical sources ? strictly better than the finder's "Drive / CF dashboard" suggestion for the AUD. (b) DEPLOY_ALPHA.md:143 is partial rot, not total: the SPA does degrade gracefully when the engine is unreachable (SetupWizard offline branch, web/src/components/SetupWizard.tsx:84-142), but via a capabilities fetch, and there is no persistent "engine offline" banner and zero /api/health references ? so soften rather than delete the claim. Also worth citing commit 5ffb750 / PR #117 and the "Cloudflare is now the single canonical host" line in the doc PR as the retirement record, since no .md currently captures it.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- verifier reasoning: Personally reproduced every element of the claim in the current working tree (khelsutra @ f9eea14). (1) deploy/DEPLOY_ALPHA.md:3 reads "***Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22...)" and :35 names "the CPU droplet `168.144.126.176`" as the engine box; :143 claims the SPA "shows a friendly 'engine offline' banner (`/api/health` ping)". (2) Commit 5ffb750 (2026-06-25, PR #117) states verbatim: "That droplet has been decommissioned (its data archived to gs://khelsutra-rally-corporus/archive/droplet-claude-do-rally-test-20260625/)... Cloudflare is now the single canonical host" ? it only removed .github/workflows/parity.yml + site/scripts/parity-check.sh, touching neither deploy doc. (3) grep -i 'retired|decommission' over all *.md in the repo = 0 matches, so no doc records the retirement. (4) deploy/GPU_VM_SERVING.md:24 still says "the droplet stays as a hot fallback"; :37-40 instruct sourcing the CF-Access AUD from "the droplet's /srv/khelsutra/config.json" and the tunnel token from "systemctl cat cloudflared" on the droplet ? and that file was last edited 2026-06-29 (ca8bfcc #119), FOUR DAYS AFTER the decommission, without fixing this, so NOT already-fixed. (5) grep over web/src: zero references to /api/health anywhere (only FfmpegBanner.tsx exists as a banner; SetupWizard.tsx:84-142 has an offline message driven by a capabilities fetch, not a /api/health ping), confirming the :143 rot. (6) gh: issues #114 (allowlisted email couldn't sign in, real demo 2026-06-23) and #97 (split-origin breaks CF Access ? single-origin needed) are both OPEN, contradicting "Status: LIVE ... only a rotated Gemini key + a browser OTP login remain". Concrete failure: this is a docs-as-memory, multi-session project ? the next serving session following GPU_VM_SERVING.md \$0 is blocked at prerequisites trying to SSH a nonexistent box for secrets, and DEPLOY_ALPHA.md's LIVE status + "hot fallback" cutover-safety claim would drive wrong go-live/rollback decisions for the actual alpha. P1 is honest for the primary serving runbook; effort S (one doc PR) is right. The doc PR itself is not owner-gated; only the Cloudflare-dashboard CNAME/Access cleanup is, and the proposed fix already scopes that as a note.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- verifier reasoning: web/src/components/CompileResult.tsx:76-77 carries the comment "wa.me can't attach the local file, so the user adds the just-downloaded reel", and :78-79 builds the text-only link `https://wa.me/?text=\${encodeURIComponent(t("result.shareText"))}` ? exactly as claimed. Grep for navigator.share / navigator.canShare / webShare (case-insensitive) across web/ returns zero

matches, so no Web Share API attempt exists anywhere in the SPA. git log for the file shows the newest commit touching it is 0fae364 "feat(web+site): A2 ? localized WhatsApp share for the reel (#82)" ? the commit that ADDED the wa.me link; nothing since fixed it (working tree clean on master...origin/master). gh issue view 112 (avidullu/khelsutra): state OPEN, labeled "bug", body matches the finding including the CF-Access rationale. That rationale is corroborated in-repo: deploy/GPU_VM_SERVING.md:37-38 and :87 show serving sits behind Cloudflare Access (cf_access_aud / cf_team_domain, "auth = the Cf-Access JWT verify"), so a shared reel URL is unreachable for recipients outside the tenant ? the file-share path is the only real fix. Not owner-gated: contained frontend UX change implementing the owner's own written plan in #112; no go-live, corpus/training, spend, or legal surface.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:- [P2] khelsutra (avidullu/khelsutra): ****Issue #114 stale-open: CF Access guest-login debugging against a deployment that no longer exists**** ? Debt is real but the fix is an owner tracker call on the owner-gated alpha/serving track. Verified: (1) gh issue view 114 ? OPEN, createdAt == updatedAt == 2026-06-23T10:25:52Z, 0 comments; body is CF-dashboard-only triage of the Access allowlist for the droplet alpha and explicitly says "Deferred by owner 2026-06-23". (2) Cited anchor exists: deploy/DEPLOY_ALPHA.md:106 is the Allow-policy/Emails-allowlist step (line 105 is the app-domain step ? off by one). (3) Deployment gone: memory session-handoff.md:19 ? droplet claude-do-rally-test/168.144.126.176 deleted 2026-06-25, "the prior 'LIVE alpha on the droplet' ... is GONE ? alpha is paused"; api.khelsutra.guru CNAME/CF-Access flagged for cleanup. (4) Not already fixed: git log --deploy/DEPLOY_ALPHA.md shows nothing relevant after 2026-06-23 (latest 6e4dbaf PR #113 is the unrelated GPU-VM doc); no in-repo reference to #114. (5) Nothing in-repo can fix or verify it ? the Access policy lives in the Cloudflare dashboard and the origin behind app./api. no longer exists. Fix requires owner disposition (close vs retitle), squarely under the alpha/serving go-live gate.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:| Product backlog | Non-audit feature issues remain open (for example live sharing/access, richer video-info panel, quality/perf/backlog tickets). | Normal product prioritization, not part of this audit remediation board. |

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/BACKLOG.md:- ? ****One-shot "serving startup-script" for the GPU VM**** ? `create\_gpu\_vm.sh` already injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs the ****Ops Agent****; standing up the live engine on the VM (`khelsutra/deploy/GPU\_VM\_SERVING.md`) is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs ? pulls the engine source (from GCS) + builds the venv ? *(optional)* the WASB native env (`gpu\_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs `cloudflared` with the tunnel token ? starts `khelsutra-engine`. So a GPU serving box comes up ****hands-off**** (reuses the `deploy/gcp/nightly\_regression/run\_on\_vm.sh` bootstrap pattern). ****Why deferred ? the design hinge is SECRETS:**** the Gemini key / CF-Access AUD / tunnel token must ****not**** sit in a plaintext startup-script (instance metadata is broadly readable by anyone with `instances.get`); they should come from ****Secret Manager**** (VM SA ? `secretAccessor`) or be injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs fetched ? not just a script (owner asked 2026-06-23). ****Revisit:**** when the GPU-VM serving path is used often enough to automate.

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/COMPUTE_DECOUPLED_SERVING/README.md:| M9 | ****Edge auth**** (CF-Access extraction) + per-tenant scoping on `user\_id`; DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | [#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; ****+ exposure-safety boot posture**** in `startup.py` ? edge-auth-on requires the `allowed\_video\_root` jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ? owner/counsel) |

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/COMPUTE_DECOUPLED_SERVING/README.md:- 2026-06-21 ? ****M9-auth code landed**** ([#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)): `backend/api/auth.py` ? Cloudflare Access ****Cf-Access JWT verify**** (RS256: sig+exp+iss+aud against the team JWKS; lazy `PyJWT[crypto]`; injectable JWKS ? fully-offline tests) + `resolve\_tenant` ? tags `create\_job(owner\_id=?)`; `SecurityConfig.{edge\_auth\_enabled=False,cf\_team\_domain,cf\_access\_aud}` + CORS via `RALLY\_CORS\_ALLOW\_ORIGINS`. ****Default-OFF.**** Security review (`wf\_9b4c71d0`) found NO bypass; locked alg=none + HS256-confusion rejection. ****? M9 ? ? the consent cols + ToS/DPA are owner/counsel; the consent migration is now v7 (M8 took v6).****

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/COMPUTE_DECOUPLED_SERVING/README.md:- 2026-06-21 ? **M9** hardening: exposure-safety boot posture** (alpha-shareable "tunnel-to-a-box" enabler): extended `backend/config/startup.py` with a **server-only exposure posture** (new `include\_exposure=True` from `\_enforce\_startup\_or\_exit`; the worker stays unaffected ? it reads `--video-uri`, not untrusted tenant paths). Gated on `security.edge\_auth\_enabled` (**default-OFF** ? strict no-op**, byte-identical to today): when the owner flips edge auth ON to expose the engine behind the CF-Access gate, three checks fire at **boot** ? `EXPOSED_NO_VIDEO_ROOT` (**fatal**: `allowed_video_root` jail unset ? a tenant could read an arbitrary local file), `EXPOSED_EDGE_AUTH_INCOMPLETE` (**fatal**: CF team-domain/AUD missing ? lifts the current first-request 401 to boot), `EXPOSED_AUTH_EXTRA_MISSING` (**warn**: `PyJWT[crypto]` not importable ? would 500 per request). So a half-configured exposure fails fast instead of leaking/500-ing. +15 unit tests (default-OFF no-op, worker-ignores, both-fatals-compose-with-profile-checks, server-exit, WARN-only-no-exit). Adversarial 3-lens review (`wf_35b735a9`) confirmed default-OFF byte-identical + no bypass; folded a **whitespace-drift** (aligned `auth.resolve_tenant` to `.strip()` team/AUD so "configured" means the same at boot + at request time) + named the config fields in the edge-auth-incomplete message. Pairs with the appliance **T3** network posture (raw `:8000` refused ? owner-run). Suite green, ruff+mypy clean. **backstops M9** ? the actual exposure flip + CF team/AUD/ingress + key rotation stay owner-side. (Deferred to the PR-2 runbook: document that `upload_dir` must sit under `allowed_video_root`, and that you **MUST** set `edge_auth_enabled=True` when exposing ? these checks only fire then.)**

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/HOSTING_PLAN.md: ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/HOSTING_PLAN.md:browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/HOSTING_PLAN.md:Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ? rate

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/docs/UI_ALPHA_EXECUTION_PLAN.md:- [] **D2** (Claude):** DNS records; `cloudflared` Windows service on the box ? `api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env

42. assistant (2026-07-04T02:26:58.947Z)

Perfect! Now let me compile all the findings into a comprehensive summary. Let me also verify the Dockerfile does NOT currently install the auth extra:

43. user (2026-07-04T02:27:00.312Z)

55:# Install with the [storage-gcs] extra: the cloud worker PULLS the gs:// input and PUSHES the
65: && python3 -m pip install "[storage-gcs]"

44. assistant (2026-07-04T02:28:03.124Z)

Perfect. Now let me create a comprehensive summary of the findings:

****Cloudflare Access Edge Auth (M9) ? Exact Lockout-Safe Map****

**** (A) REQUIRED CONFIG JSON FOR EDGE AUTH ENABLEMENT ****

```
```json
{
 "security": {
 "edge_auth_enabled": true,
 "cf_team_domain": "https://<team>.cloudflareaccess.com",
 "cf_access_aud": "<application-audit-tag-from-cf-dashboard>",
 "allowed_video_root": "/var/lib/videos"
 }
}
```

**\*\*All required fields and their defaults (backend/config/models.py:358?389):\*\***

Field	Type	Default	Notes
`edge_auth_enabled`	bool	`False`	<b>**DEFAULT-OFF**</b> ? no JWT required/verified (local single-user unchanged); when True, `/api/process`, `/api/assets`, `/api/labels` REQUIRE + verify Cf-Access JWT
`cf_team_domain`	str	``	e.g. `https://myteam.cloudflareaccess.com` (issuer + JWKS source: `{domain}/cdn-cgi/access/certs`)
`cf_access_aud`	str	``	The AUD tag of the CF Access <b>**application**</b> (extracted from Cloudflare dashboard under the app's policy settings)
`allowed_video_root`	Optional[str]	`None`	<b>**LOAD-BEARING**</b> for exposure: a jail directory that all `/api/process` video paths must resolve under; <b>**unset + edge_auth_enabled=True = fatal boot error**</b>
`upload_dir`	str	`"output/uploads"`	Must sit under `allowed_video_root` in multi-tenant mode (else a tenant uploads can be read back arbitrarily)
`max_upload_mb`	int	`4096`	Whole-file cap (4 GB default)

---

## **\*\* (B) JWT VERIFICATION IMPLEMENTATION & HEADER HANDLING \*\***

**\*\*File: `backend/api/auth.py`\*\***

**\*\*Request header read (line 23, 140):\*\***

- Header name: **\*\*`Cf-Access-Jwt-Assertion`\*\*** (case-insensitive via Starlette; also tries lowercase variant for plain dicts in tests)
- Missing header + `edge\_auth\_enabled=True` ? raises `EdgeAuthError` (? 401 in HTTP routes)

**\*\*Verification details (lines 69?111):\*\***

- **\*\*Algorithm:\*\*** RS256 only (hardcoded in `jwt.decode(..., algorithms=["RS256"], ...)` line 93)
- **\*\*JWKS fetch:\*\*** `\_default\_jwks\_fetcher` (line 44) ? `{team\_domain}/cdn-cgi/access/certs`
- **\*\*Required claims checked (line 96):\*\*** `exp`, `iss`, `aud` (Cloudflare enforces all three)
- **\*\*Signature verification:\*\*** RSA public key from JWKS (line 89)
- **\*\*Issuer check (line 95):\*\*** must equal `team\_domain.rstrip("/")`
- **\*\*Audience check (line 94):\*\*** must equal the configured `cf\_access\_aud`
- **\*\*User identity extraction (line 107):\*\*** returns `claims.get("email")` if present, else `claims.get("sub")`
- **\*\*JWKS caching:\*\*** 10-minute TTL (line 25?26); injectable for tests

**\*\*PyJWT import (lines 75?83):\*\***

```
```python
import jwt # LAZY-IMPORTED ? only when resolve_tenant is called with edge_auth_enabled=True
from jwt import PyJWKSet
```
```

? **\*\*If PyJWT[crypto] is not installed, every request raises `EdgeAuthError`\*\*** with message:  
 `"PyJWT[crypto] is required for edge auth ? install the 'auth' extra ?"`

**\*\*Entry point ? resolver function (lines 113?148):\*\***

```
```python
def resolve_tenant(headers, config, *, jwks_fetcher=None) -> Optional[str]:
    # Returns None when edge_auth_enabled=False (no-op)
    # Raises EdgeAuthError when enabled + missing/invalid token
    # Returns verified user_id (email or sub) when valid
```
```

**\*\*HTTP route integration (backend/main.py:671, 743, 904, 1084):\*\***

- `/api/labels` (line 671): edge-auth mirror; 401 on failure
- `/api/assets` (line 743): tags `owner\_id=?` when creating an asset
- `/api/process` (line 904): tags `owner\_id=?` when creating a rally job
- `/api/compile` (line 1084): tags `owner\_id=?` on compile jobs

**\*\*EXEMPT endpoint:\*\***

- **\*\*`/api/health` (line 314):\*\*** NO auth required ? used by Cloud Run health probes + browser offline detection

---



## **\*\* (C) BOOT-TIME SAFETY CHECKS & CRASH-PREVENTION \*\***

**\*\*File: `backend/config/startup.py:93?168`\*\*** (exposure-safety posture for exposed HTTP servers)

**\*\*When enabled (edge\_auth\_enabled=True), the server runs three FATAL checks at startup:\*\***

```
Check Code	Level	Trigger	Resolution
`EXPOSED_NO_VIDEO_ROOT`	**FATAL**	`allowed_video_root` unset/empty/whitespace-only	Set
`security.allowed_video_root` to a jail directory (e.g., `/var/lib/videos`). An unauthenticated			
tenant could otherwise read `../../etc/passwd` from the local filesystem.			
`EXPOSED_EDGE_AUTH_INCOMPLETE`	**FATAL**	`cf_team_domain` OR `cf_access_aud`	
unset/empty/whitespace-only (lines 124?139)	Set BOTH: `cf_team_domain` (e.g.,		
`https://myteam.cloudflareaccess.com`) and `cf_access_aud` (the app's audience tag from CF			
dashboard). Without these, every JWT verify fails ? 401 on first request; boot-check lifts this			
to startup.			
`EXPOSED_AUTH_EXTRA_MISSING`	**WARN** (non-fatal)	`PyJWT[crypto]` not importable (lines	
144?152)	Install the `auth` extra: `pip install -e .[auth]`. Warning only because the auth		
module's own ImportError at runtime is the authority. ? **But every request will 500** if PyJWT			
is missing.			
`EXPOSED_HOST_ALLOWLIST_LOOPBACK`	**WARN** (non-fatal)	`RALLY_ALLOWED_HOSTS` env unset AND	
edge_auth_enabled=True (lines 157?166) | Set `RALLY_ALLOWED_HOSTS` env var to your public
domain(s) or `` (if the edge gate validates host). Default loopback-only guard will 400-reject
the proxied requests. |
```

**\*\*Server startup enforcement (backend/main.py:1206?1218):\*\***

```
```python
def _enforce_startup_or_exit(config):
    try:
        enforce_startup_checks(config, logger=logger, include_exposure=True)
    except StartupCheckError as e:
        logger.error("[STARTUP] refusing to start: %s", e)
        sys.exit(1) # ? CRASH-LOOP if both FAYALS are unset
...

```

****Called at server startup (backend/main.py line 1396)**** before FastAPI `app` is created.

****Whitespace semantics (line 131?132, 124):****

All config values are `.strip()`'d; empty string or whitespace-only = unset.

**** (D) DEPLOYED IMAGE & PyJWT[crypto] STATUS ****

****File: `deploy/cloudrun/Dockerfile`****

****Current install (line 65):****

```
```dockerfile
pip install "[storage-gcs]"
...

```

? **\*\*PyJWT[crypto] is NOT installed\*\*** ? the image has `storage-gcs` but NOT the `auth` extra.

**\*\*PyJWT is declared in (pyproject.toml:47?60):\*\***

```
```toml
[project.optional-dependencies]
dev = ["pytest", "pytest-cov", "httpx", "ruff", "mypy", "PyJWT[crypto]>=2.8"] # CI + tests only
auth = ["PyJWT[crypto]>=2.8"] # M9 edge-auth
runtime
...

```

****Fix for deployed image (Dockerfile line 65 ? MUST update before turning ON edge_auth):****

```
```dockerfile
```

**Change this:**

```
RUN python3 -m pip install ".[storage-gcs]"
```

## To this (for edge auth support):

```
RUN python3 -m pip install ".[storage-gcs,auth]"
```
```

****Impact of NOT rebuilding the image:****

- If ``edge_auth_enabled=True`` is set in config ? boot-check logs ****WARN**** ``EXPOSED_AUTH_EXTRA_MISSING``
- First ``/api/process`` request ? ``EdgeAuthError: "PyJWT[crypto] is required?"`` ? 500 Internal Server Error
- ****Symptom:**** server starts but all authenticated requests fail with 500

**** (E) BIND HOST & INGRESS INTERACTION ****

****File: `backend/server\_runtime.py:46?61`**** (``_resolve_bind_host`` function)

****Precedence:****

1. CLI ``--host`` flag (highest)
2. ``RALLY_BIND_HOST`` env var
3. Auto-detect based on ****`edge\_auth\_enabled`**** (line 59?60):
 - If ``edge_auth_enabled=True`` ? bind ``0.0.0.0`` (all interfaces, behind CF proxy)
 - If ``edge_auth_enabled=False`` (default) ? bind ``127.0.0.1`` (localhost only, safe default)

****Your current deployment already passes ``--host 0.0.0.0`` explicitly ? no change needed.****

The edge-auth check is a ****safety fallback**** for auto-detection when no explicit flag is given.

**** (F) INGRESS LOCK + DNS / DOMAIN MAPPING PROCEDURE ****

****Documented in: `docs/HOSTING\_PLAN.md` + `deploy/cloudrun/README.md`****

****For Cloud Run control-plane service (rally-control-plane, asia-southeast1):****

****1. Lock Cloud Run ingress to Cloudflare proxy only:****

```
```bash
gcloud run services update rally-control-plane \
 --region asia-southeast1 \
 --ingress=internal-and-cloud-armor \
 --set-cloud-armor-policy=cf-only
```
```

(Alternatively, use Cloud Armor + GCP's Cloudflare IP allowlist ? consult GCP docs for exact policy syntax. The exact command is security-dependent and not fully in docs yet.)

****2. Create/update Cloud Run domain mapping to `api.khelsutra.guru``:**

```
```bash
gcloud run domain-mappings create \
 --service=rally-control-plane \
 --domain=api.khelsutra.guru \
 --region=asia-southeast1
```

or update:
```bash
gcloud run domain-mappings update api.khelsutra.guru \
 --service=rally-control-plane \
 --region=asia-southeast1
```
```

This returns a ****Cloud Run-assigned CNAME**** (e.g., ``ghs.googleusercontent.com/?``).

****3. Set Cloudflare DNS CNAME:****

In Cloudflare dashboard:

- Type: `CNAME`
- Name: `api.khelsutra.guru` (or `api` if the zone is `khelsutra.guru`)
- Target: the Cloud Run CNAME from step 2
- ****Proxied**** (orange cloud icon) ? so traffic flows through CF's edge

****4. Create/configure Cloudflare Access application:****

- Name: "KhelSutra API (rally-control-plane)"
- Domain(s): `api.khelsutra.guru`
- ****Copy the AUD tag**** from the app's general settings ? paste into `config.json` as `cf\_access\_aud`
- Add Access policy (e.g., allow emails matching a list or a specific domain)

****5. Export CF team domain & test routing:****

```
`bash
CF_TEAM_DOMAIN="https://your-team.cloudflareaccess.com"
curl -i "https://api.khelsutra.guru/api/health" \
  -H "Cf-Access-Jwt-Assertion: <test-jwt>"
`
```

****Status in docs:****

- `docs/HOSTING\_PLAN.md:14?32` sketches the topology
- `deploy/cloudrun/README.md:65?84` covers the control-plane service deploy with required flags
- ****No step-by-step runbook for M9 ingress-lock + DNS exists yet**** (owner-gated detail)

**** (G) EXISTING CLOUDFLARE ACCESS APP REFERENCE ****

****Documented in: `docs/HOSTING\_PLAN.md:30?42` + archived audit reports****

****Existing "KhelSutra Studio (alpha)" Cloudflare Access application (on app.khelsutra.guru):****

- ****Team domain:**** `https://<team>.cloudflareaccess.com` (same team as the new API app)
- ****Domain:**** `app.khelsutra.guru` (the React SPA, served via Cloudflare Pages)
- ****Auth:**** Cloudflare Access email-OTP allowlist (or OIDC)

****Replication notes for api.khelsutra.guru:****

- Use the ****same team domain**** (no new team needed)
- Create a ****separate Access application**** scoped to `api.khelsutra.guru` (one app per domain)
- Each app has its own ****AUD tag**** (e.g., `app-aud-uuid-1` for app.khelsutra.guru, `api-aud-uuid-2` for api.khelsutra.guru)
- Extract the ****api.khelsutra.guru AUD**** from CF dashboard ? add to backend config as `cf\_access\_aud`
- Use the ****same team domain**** for `cf\_team\_domain` in both app and API configs

****Archive note (2026-07-02 audit):****

- The original alpha on the DigitalOcean droplet (168.144.126.176) was ****retired 2026-06-25**** (data archived to `gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`)
- Cloudflare Access setup for the alpha is recoverable from the archive if needed
- The current alpha is ****paused**** pending Cloud Run deployment (api.khelsutra.guru routing to be configured)

****SUMMARY TABLE: DEPLOYMENT CHECKLIST****

| Step | Item | File:Line | Status | Lockout Risk |
|------|---|---------------------------------|--|---|
| 0 | Install `auth` extra in Dockerfile | `deploy/cloudrun/Dockerfile:65` | **REQUIRED before image rebuild** | ? CRASH: 500 on every auth request if missing |
| 1 | Set `security.edge_auth_enabled = true` | `config.json` | USER choice | ? boot check catches it |
| 2 | Set `cf_team_domain` (CF team domain) | `config.json` | **REQUIRED** | ? FATAL boot error |

```

if missing |
3	Set `cf_access_aud` (app AUD from CF dashboard)	`config.json`	**REQUIRED**	? FATAL boot error if missing
4	Set `allowed_video_root` (jail directory)	`config.json`	**REQUIRED**	? FATAL boot error if missing
5	Set `RALLY_ALLOWED_HOSTS` env var	deployment env	**REQUIRED**	? WARN at boot (400 on forwarded requests)
6	Cloud Run domain-mapping to `api.khelsutra.guru`	gcloud CLI	**REQUIRED**	? DNS fails if not set
7	Cloudflare DNS CNAME ? Cloud Run	CF dashboard	**REQUIRED**	? DNS fails if not set
8	Create Cloudflare Access app for `api.khelsutra.guru`	CF dashboard	**REQUIRED**	? Access blocks (401) if missing
9	Extract Access app AUD ? config	CF dashboard ? config	**REQUIRED**	? FATAL boot error if missing
10	Set Cloud Run ingress to CF-only	gcloud CLI	**STRONGLY RECOMMENDED**	? Direct Cloud Run URL bypasses Access

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****KEY BOOT-CRASH TRAPS****

1. ****Image without `[auth]` extra (Dockerfile NOT updated)**** ? WARN at startup but ****500** on first request ? looks like a deployment bug, not auth config
2. ****`edge\_auth\_enabled=True` but missing `allowed\_video\_root`**** ? ****FATAL: sys.exit(1)**** before server starts ? crash-loop on restart
3. ****`edge\_auth\_enabled=True` but `cf\_team\_domain` or `cf\_access\_aud` empty/unset**** ? ****FATAL: sys.exit(1)**** ? crash-loop
4. ****`RALLY\_ALLOWED\_HOSTS` unset + edge auth enabled**** ? ****WARN at boot**** but 400-rejects all proxied requests ? looks like a routing bug
5. ****Cloud Run domain-mapping OR Cloudflare CNAME not configured**** ? DNS fails to resolve `api.khelsutra.guru` ? user can't reach the API (not a crash, but inaccessible)

****EXACT EVIDENCE (file:line citations)****

| Question | Evidence | File:Line |
|---|---|-------------------------------------|
| ----- ----- ----- | | |
| **1. CF Access JWT header name** | `Cf-Access-Jwt-Assertion` | `backend/api/auth.py:23` |
| **1. Header read (case-insensitive)** | Two variants tried (uppercase + lowercase) | |
| | `backend/api/auth.py:140` | |
| **1. RS256 verification** | `algorithms=["RS256"]` only | `backend/api/auth.py:93` |
| **1. JWKS URL source** | `{team_domain}/cdn-cgi/access/certs` | `backend/api/auth.py:47` |
| **1. Audience check** | `jwt.decode(?, audience=aud, ...)` | `backend/api/auth.py:94` |
| **1. Issuer check** | `issuer=issuer(team_domain)` | `backend/api/auth.py:95` |
| **1. User identity extraction** | `email` if present else `sub` | `backend/api/auth.py:107` |
| **1. PyJWT lazy import** | `import jwt` inside function | `backend/api/auth.py:75?76` |
| **1. Missing header status** | 401 (HTTPException via route wrapper) | |
| | `backend/main.py:904?906` | |
| **1. Routes gated (sample)** | `/api/process`, `/api/assets`, `/api/labels` | |
| | `backend/main.py:904, 743, 671` | |
| **1. Health exempt** | `/api/health` has NO auth check | `backend/main.py:314?326` |
| **2. Config field names** | `edge_auth_enabled`, `cf_team_domain`, `cf_access_aud`,
`allowed_video_root` | `backend/config/models.py:358?389` |
| **2. Defaults** | `edge_auth_enabled=False`, cf fields empty string, `allowed_video_root=None` | `backend/config/models.py:381?385` |
| **2. Boot-time checks** | `_exposure_checks()` function | `backend/config/startup.py:93?168` |
| **2. FATAL check 1 (no root)** | `EXPOSED_NO_VIDEO_ROOT` | `backend/config/startup.py:109?119` |
| | | |
| **2. FATAL check 2 (incomplete team/aud)** | `EXPOSED_EDGE_AUTH_INCOMPLETE` | |
| | `backend/config/startup.py:124?139` | |
| **2. WARN check (missing auth extra)** | `EXPOSED_AUTH_EXTRA_MISSING` | |
| | `backend/config/startup.py:144?152` | |
| **2. WARN check (host allowlist)** | `EXPOSED_HOST_ALLOWLIST_LOOPBACK` | |

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`backend/config/startup.py:157?166` |  
| **2. Server exit on FATAL** | `sys.exit(1)` via `_enforce_startup_or_exit` |  
`backend/main.py:1217?1218` |  
| **3. Bind host precedence** | edge_auth ? `0.0.0.0`, else `127.0.0.1` |  
`backend/server_runtime.py:59?61` |  
| **4. PyJWT[crypto] in pyproject.toml** | `auth = ["PyJWT[crypto]>=2.8"]` | `pyproject.toml:60`  
|  
| **4. Current Dockerfile** | `pip install "[storage-gcs]"` (no auth) |  
`deploy/cloudrun/Dockerfile:65` |  
| **5. Ingress/DNS procedure** | Cloud Run domain-mapping + CF CNAME + Access app |  
`deploy/cloudrun/README.md:65?84`, `docs/HOSTING_PLAN.md:30?42` |  
| **6. Existing Access app** | app.khelsutra.guru (SPA); same team as API |  
`docs/HOSTING_PLAN.md:30?42` |
```